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CHEMISTRY

Class – 10

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1. Periodic Classification of Elements

Solutions

Worksheet – 1

1. **Correct Answer: C**
   Dmitri Mendeleev formulated the Periodic Law, which stated that the properties of elements are a periodic function of their atomic masses (or atomic weights).
2. **Correct Answer: C**
   When Mendeleev began his work on the periodic classification of elements in 1869, only 63 elements had been discovered.
3. **Correct Answer: B**
   In periodic table terminology, the vertical columns that contain elements with similar chemical properties are called Groups.
4. **Correct Answer: B**
   The horizontal rows are known as periods, while the vertical columns are known as groups.
5. **Correct Answer: C**
   In a triad, the atomic mass of the middle element is approximately the arithmetic mean of the other two elements.
6. **Correct Answer: A**
   Similarly, Mendeleev predicted an element below Aluminum in the periodic table and called it Eka-Aluminum. This element was later discovered and named Gallium (Ga).
7. **Correct Answer: C**
   Eka-Silicon was the provisional name given by Mendeleev to the element predicted to be below silicon. It was later discovered and named Germanium (Ge).
8. **Correct Answer: C**
   The 1st period is the shortest, containing only 2 elements (Hydrogen and Helium). The 6th period is considered the longest as it contains 32 elements (including the Lanthanides).
9. **Correct Answer: C**
   2nd Period: 8 elements (Li to Ne). 3rd Period: 8 elements (Na to Ar). 4th Period: 18 elements (K to Kr). 5th Period: 18 elements (Rb to Xe).
10. **Correct Answer: A**
    The Modern Periodic Law states that the physical and chemical properties of elements are periodic functions of their atomic numbers, whereas Mendeleev's law was based on atomic mass.

Worksheet – 2

1. **Correct Answer: B**
   The alkaline earth metals are the elements in Group II A (Group 2) of the periodic table. They include Beryllium, Magnesium, Calcium, Strontium, Barium, and Radium.
2. **Correct Answer: A**
   The transition elements (d-block elements) start from the 4th Period. The first transition series begins with Scandium (Sc), which has the atomic number 21.
3. **Correct Answer: D**
   The term "rare earth metals" or "rare earth elements" specifically refers to the 15 elements in the Lanthanide series (plus scandium and yttrium).
4. **Correct Answer: D**
   Helium (He) has a valence shell configuration of 1s2 (2 electrons), known as a duplet. All other inert (noble) gases have an octet configuration (ns2 np6) with 8 valence electrons.
5. **Correct Answer: B**
   Lead (Pb) is a p-block element, often classified as a post-transition metal.

6. **Correct Answer: B**

Metalloids (such as Boron, Silicon, Germanium, Arsenic, Antimony, Tellurium, and Polonium) are found along the "staircase" line that separates metals from non-metals. This region is located entirely within the p-block of the periodic table.

7. **Correct Answer: C**

The "differentiating electron" is the last electron added to an atom. A d-subshell consists of 5 orbitals, each capable of holding 2 electrons. Therefore, the maximum capacity of a d-subshell is 5 × 2 = 10 electrons

8. **Correct Answer: A**

The general formula for d-block elements is typically written as $(n-1)d^{1-10}ns^{0-2}$.

9. **Correct Answer: A**

The general electronic configuration for f-block elements is typically written as $ns^2(n-1)d^{0or1}(n-2)f^{1-14}$

10. **Correct Answer: C**

Lanthanum (Atomic Number 57) has the electronic configuration $[Xe]5d^16s^2$. Although it gives its name to the Lanthanide series, its last (differentiating) electron actually enters the 5d subshell, not the 4f subshell.

Worksheet – 3

1. **Correct Answer: A**

As you move across a period from left to right, the atomic number increases, meaning more protons are added to the nucleus. Since electrons are added to the same valence shell, the increased effective nuclear charge pulls the electrons closer to the nucleus, causing the atomic radius to decrease.

2. **Correct Answer: A**

Metallic character refers to the tendency of an atom to lose electrons. As you move across a period, the nuclear attraction increases, making it harder for atoms to lose electrons. Therefore, elements become less metallic and more non-metallic.

3. **Correct Answer: A**

Atomic size decreases as you move from left to right.

4. **Correct Answer: C**

Atomic radius increases down a group and decreases across a period.

5. **Correct Answer: D**

The atomic radius decreases as you move from left to right across a period due to increased nuclear attraction.

6. **Correct Answer: D**

Cations are always smaller than their parent neutral atoms because losing electrons increases the effective nuclear charge on the remaining electrons, pulling them closer to the nucleus.

7. **Correct Answer: C**

Ionization potential (or Ionization Energy) is defined as the minimum energy required to remove the most loosely bound electron from an isolated gaseous atom (X(g)) to form a gaseous cation ($X^+(g)$).

8. **Correct Answer: B**

Fluorine (F) is the most electronegative element. Oxygen (O) is the second most electronegative element with a value of approximately 3.44.

9. **Correct Answer: C**

Halogens have the highest effective nuclear charge in their respective periods and need only one electron to complete their stable octet. Therefore, they release the most energy (highest electron affinity) when accepting an electron.

10. **Correct Answer: B**

Electronegativity increases across a period (left to right) and decreases down a group.

Competitive corner

SINGLE CORRECT OPTION TYPE

1. **Correct Answer: B**
   Elements in the same group share similar chemical properties and the same number of valence electrons. Magnesium (Mg) and Strontium (Sr) both belong to Group 2 (Alkaline Earth Metals).
2. **Correct Answer: C**
   The defining characteristic of a group (vertical column) is that all elements in it have the same number of electrons in their outermost shell (valence shell). For example, all Group 1 elements have 1 valence electron.
3. **Correct Answer: C**
   The octet configuration refers to having a stable outer shell of 8 electrons. This is characteristic of the noble gases, which are placed in Group 18 of the modern periodic table.
4. **Correct Answer: C**
   The length of a period is determined by the total number of electrons required to fill the atomic orbitals (s, p, d, f) associated with that specific energy level. For instance, Period 1 has 2 elements because the s-subshell holds 2 electrons.
5. **Correct Answer: C**
   The modern "long form" of the periodic table consists of 7 horizontal rows called periods.
6. **Correct Answer: C**
   Although Lanthanum (La) gives its name to the Lanthanide series (f-block), it is technically a d-block element. Its electron configuration is $[Xe]5d^16s^2$ meaning the differentiating (last) electron enters the 5d subshell, not the 4f subshell.
7. **Correct Answer: D**
   The Actinides (or Actinoids) are the series of inner transition elements found in the f-block. While Actinium (Atomic Number 89) is often grouped with them, the series of 14 elements involving the filling of the 5f orbitals technically runs from Thorium (90) to Lawrencium (103).
8. **Correct Answer: B**
   Period 6: Contains s-block (Cs, Ba), d-block (La, Hf-Hg), and p-block (Tl-Rn) elements. (It also contains f-block elements).
9. **Correct Answer: B**
   The Atomic Radius is defined scientifically as the distance from the center of the nucleus to the boundary of the surrounding shells of electrons.
10. **Correct Answer: B**
    The Covalent radius is based on the distance between bonded atoms and is smaller. The van der Waals radius is based on the distance between non-bonded atoms and is significantly larger.
11. **Correct Answer: C**
    Mg (Period 3) and Ca (Period 4) are in the same group (Group 2). Atomic radius increases down a group because new electron shells are added. Therefore, Magnesium is smaller than Calcium (Mg < Ca).
12. **Correct Answer: C**
    These species are isoelectronic, meaning they all have the same number of electrons (10 electrons). In an isoelectronic series, ionic radius decreases as the nuclear charge (number of protons) increases.
13. **Correct Answer: D**
    For the same element, the size depends on the electron-electron repulsion.
14. **Correct Answer: D**
    Calcium (Z=20) loses 2 electrons to become stable.
15. **Correct Answer: D**
    Ionization energy is the energy required to remove an electron. An atom can theoretically have as many ionization energies as it has electrons. Since a neutral Carbon atom has an atomic number of 6, it has 6 electrons. Therefore, it is possible to remove all 6 electrons one by one, resulting in 6 distinct ionization energies.

16. **Correct Answer: B**
Penetrating power refers to how close an electron in an orbital can get to the nucleus. The order of penetration for subshells within the same principal quantum number is s > p > d > f. The s-orbital is spherical and provides the most electron density near the nucleus.
17. **Correct Answer: C**
Ionization energy generally decreases down a group and increases across a period.
18. **Correct Answer: D**
The zero group refers to the Noble Gases (Group 18). Elements in this group have a completely filled valence shell (stable octet configuration), making them extremely stable. Removing an electron from such a stable configuration requires the highest amount of energy in their respective periods.
19. **Correct Answer: A**
Cations are always smaller than their parent neutral atoms because they have lost electrons while retaining the same nuclear charge (protons), leading to a stronger pull on the remaining electrons.
20. **Correct Answer: C**
Metals generally have a low tendency to gain electrons (prefer losing them). Nitrogen has a stable half-filled $2p^3$ configuration, so it resists gaining an electron (low affinity). Oxygen has high tendency to gain electrons. Chlorine has highest tendency to gain electrons to complete its octet.
21. **Correct Answer: D**
Robert Mulliken proposed that electronegativity is the arithmetic mean of the ionization energy and electron affinity.
22. **Correct Answer: A**
Linus Pauling arbitrarily assigned a value of 2.1 (sometimes rounded to 2.2 in modern texts) to Hydrogen (H) to serve as a reference point.
23. **Correct Answer: A**
The difference in electronegativity values between two bonded atoms determines the nature of the bond.

ONE OR MORE THAN ONE CORRECT OPTION TYPE

24. **Correct Answer: A, B, C**
The atomic radius depends on the nature of bonding, effective nuclear charge, and nuclear charge.
25. **Correct Answer: A, B, C, D**
The radius increases because the electron cloud expands. The effective nuclear charge decreases because the increased shielding from the added electron reduces the net pull per electron. Adding an electron to the valence shell increases inter-electronic repulsion, pushing the shell outward. Gaining an electron is the definition of reduction, and the species accepting the electron acts as an oxidizing agent.
26. **Correct Answer: A, B, C, D**
The valence shell is n=3, so it is in the 3rd period. It has 5 valence electrons, placing it in Group 15 (10 + 5). Group 15 is known as the Nitrogen family. The differentiating electron enters a p-orbital (p-block), and it has 5 valence electrons.
27. **Correct Answer: A, B, C, D**
All the options represent the same length in different units
28. **Correct Answer: B, C, D**
Ionisation energy decreases because the outer electrons are further from the nucleus and more shielded, making them easier to remove. Electron affinity generally decreases (becomes less negative) because the larger atomic size reduces the nuclear attraction for an incoming electron. Electronegativity decreases as the increased atomic radius reduces the nucleus's ability to attract shared electron pairs.

29. **Correct Answer: A, B, and D**
Increased shielding reduces the attraction between the nucleus and valence electrons, leading to a larger atomic radius. Greater shielding lowers the effective pull on the outer electron, making it easier to remove. hielding directly reduces the net nuclear charge felt by valence electrons.

ASSERTION AND REASON TYPE

30. **Correct Answer: B**
The Reason simply provides a definition of the property. It does not explain the *anomaly* (why Cl > F). The correct explanation would involve the concept of electron density and repulsion in the 2p vs. 3p subshells.

31. **Correct Answer: A**
The Assertion (Argon is larger) is a direct result of the fact stated in the Reason (Argon uses the larger van der Waals radius while Chlorine uses the smaller covalent radius). Therefore, the Reason correctly explains the Assertion.

32. **Correct Answer: C**
Lanthanoid contraction is caused by the poor shielding (not effective shielding) of the 4f electrons. Because 4f orbitals are diffuse and shield the nucleus poorly, the increasing nuclear charge pulls the outer electrons inward, effectively canceling out the expected increase in size from adding a new shell.

33. **Correct Answer: A**
The reason correctly identifies the physical mechanism (increased shielding and shells outweighing nuclear charge) that causes the lower ionization energy described in the assertion.

34. **Correct Answer: D**
Elements with very high electronegativity (like fluorine and oxygen) act as strong oxidizing agents, not reducing agents, because they readily gain electrons. High electronegativity means a strong tendency to attract electrons, which is the characteristic of an oxidizing agent.

MATRIX MATCHING TYPE

35. **Correct Answer: a–S, b–R, c–P, d–Q**
13th group → Boron family, 14th group → Carbon family, 15th group → Nitrogen family, 18th group → Noble gases

36. **Correct Answer: a–Q, b–S, c–P, d–R**
s – Block $\rightarrow ns^{1-2}$, p Block $\rightarrow ns^2np^{1-6}$, d-Block $\rightarrow ns^{1 \text{ or } 2}(n-1)d^{1-10}$, f-Block $\rightarrow (n-2)f^{1-14}\ (n-1)d^{0 \text{ or } 1}\ ns^2$

COMPREHENSION TYPE

37. **Correct Answer: B**
Electronegativity increases across a period from left to right due to the increase in nuclear charge and decrease in atomic radius.

38. **Correct Answer: B**
This series consists of elements in Period 2. As you move from Carbon (Group 14) to Fluorine (Group 17), the effective nuclear charge increases.

39. **Correct Answer: B**
High electronegativity and high ionization potential (ionization energy) are both consequences of a strong nuclear pull-on electrons.

40. **Correct Answer: D**
Generally, electron affinity decreases down a group as atomic size increase.

41. **Correct Answer: B**
Neutral Oxygen is a non-metal with high electronegativity and a strong tendency to gain electrons to complete its octet, resulting in the most exothermic first electron gain enthalpy among the choices.

42. **Correct Answer: A**
As you move across a period, the nuclear charge increases and the atomic size decreases.

INTEGER TYPE

43. **Correct Answer: 18**
The vertical columns in the modern table are called groups, and there are 18 of them.
44. **Correct Answer: 32**
The sixth period is called very long because it accommodates 32 elements.
45. **Correct Answer: 14**
The actinide series corresponds to filling of the 5f subshell, which can accommodate a maximum of 14 electrons, so it includes 14 elements.

Previous Year Papers

1. **Correct Answer: C**
Ionization enthalpy generally increases across a period, there are two distinct "peaks" in Period 2. These two distinct peaks corresponds to Be and N.
2. **Correct Answer: B**
X (-349 kJ/mol) = This is a very high (exothermic) value, characteristic of Chlorine, which has the highest electron affinity in the periodic table. Z (-295 kJ/mol) = This value is lower than Chlorine but still significant, fitting Iodine (which is further down Group 17). Y (-200 kJ/mol): This is the lowest magnitude in the set, fitting Sulfur (Group 16), which generally has a lower electron affinity than the halogens of similar periods.
3. **Correct Answer: C**
Atomic radius follows two primary trends: it decreases across a period (left to right) and increases down a group.
4. **Correct Answer: B**
Electron gain enthalpy is generally negative (exothermic) for most elements because they release energy when gaining an electron.
5. **Correct Answer: A**
These elements are all in Period 3. While ionization enthalpy generally increases from left to right, there is an anomaly between Magnesium (Mg) and Aluminum (Al).
6. **Correct Answer: D**
A diagonal relationship occurs between certain elements in the second and third periods that are positioned diagonally to each other.
7. **Correct Answer: C**
To determine the group of a p-block element, we add 10 to the number of valence electrons (ns + np).
8. **Correct Answer: C**
These species are isoelectronic, meaning they have the same number of electrons (10 electrons each).
9. **Correct Answer: D**
According to Fajans' Rules, covalent character increases with a decrease in cation size and an increase in cation charge (high polarizing power).
10. **Correct Answer: A**
As you move down a group in the periodic table, the atomic radius increases.

2. Metals and non-metals

Solutions

Worksheet – 1

1. **Correct Answer: C) Malleability**
   Explanation: Malleability is the ability of a material to deform under compressive stress (like hammering) into thin sheets.
2. **Correct Answer: C) Graphite**
   Explanation: Graphite is an allotrope of Carbon (a non-metal). Unlike most non-metals, each carbon atom in graphite is bonded to only three others, leaving one "delocalized" electron free to move. This mobility allows graphite to conduct electricity efficiently.
3. **Correct Answer: C) Al (Aluminium)**
   Explanation: P (Phosphorus), C (Carbon), and Cl (Chlorine) are non-metals, which are generally brittle and break when stretched. Al (Aluminium) is a metal, and like most metals, it can be drawn into wires.
4. **Correct Answer: C) Both 'A' and 'B'**
   Explanation: Metals have a "sea of delocalized electrons." These free electrons can carry both thermal energy (heat) and electrical charge through the metal lattice very quickly.
5. **Correct Answer: B) Very low.**
   Explanation: Non-metals usually have low melting points because they are mostly covalent in nature. The forces of attraction between their particles (intermolecular forces) are weak, so less energy is required to break them.
6. **Correct Answer: C) Form acidic oxides**
   Explanation: This is a key chemical difference:
   - Metals typically form Basic Oxides (e.g., Magnesium Oxide).
   - Non-metals typically form Acidic Oxides (e.g., Carbon Dioxide or Sulphur Dioxide).
7. **Correct Answer: B) Homogeneous mixture**
   Explanation: Brass is an alloy made of Copper and Zinc. It is considered a homogeneous mixture because the atoms of the two metals are distributed uniformly at the molecular level, but they are not chemically bonded into a new compound.
8. **Correct Answer: Graphite**
   Explanation: Despite the name "lead," pencils actually contain a mixture of Graphite and clay. Graphite's layered structure allows layers to slide off onto paper, leaving a mark.
9. **Correct Answer: D) Carbon (Diamond)**
   Explanation: Diamond is another allotrope of carbon. In a diamond, every carbon atom is bonded to four others in a rigid, 3D tetrahedral structure. This makes it the hardest known natural substance.
10. **Correct Answer: A) Carbon**
    Explanation: While most alloys are mixtures of two or more metals, Carbon is a non-metal frequently used to create alloys. For example, Steel is an alloy of Iron and a small amount of Carbon, which makes the iron much stronger and harder.

Worksheet-2

1. **Correct Answer: C) Hydrogen**

   Explanation:
   Metals above hydrogen in the reactivity series can react with acids to displace hydrogen gas.

2. **Correct Answer: B) Acidic**

   Explanation:
   Non-metal oxides usually react with water to form acids

3. **Correct Answer: C) Basic**

   Explanation:
   Metal oxides (especially of metals from groups 1 & 2) are mostly basic and react with acids to form salts:

4. **Correct Answer: C) Hydrogen**

   Explanation:
   As in Q1, metals like Zn, Mg react with acids to release hydrogen gas:

5. **Correct Answer: B) Cu**

   Explanation:
   Copper is less reactive than hydrogen, so it does not react with dilute acids to produce hydrogen. Fe, Mg, and Zn all react because they are more reactive than hydrogen.

6. **Correct Answer: C) Acids**

   Explanation:
   Most non-metal oxides are acidic in nature. When these oxides dissolve in water, they form acids.

7. **Correct Answer: B) Sodium**

   Explanation: Highly reactive metals like sodium (Na) and potassium (K) react with cold water to form metal hydroxides and hydrogen gas. Less reactive metals require hot water or steam.

8. **Correct Answer: B) K > Na > Ca > Mg**

   Explanation: Potassium is the most reactive and reacts vigorously even with cold water, followed by Na, Ca, and Mg.

9. **Correct Answer: B) $CO_2 + H_2O \rightarrow H_2CO_3$**

   Explanation: Carbonic acid (H_2CO_3) is formed when carbon dioxide dissolves in water.

   - CO_2 is a non-metal oxide.
   - When non-metal oxides dissolve in water, they generally form acids.

10. **Correct Answer: A) Blue litmus turns red**

    Explanation: Acids turn blue litmus paper red. Non-metal oxides dissolved in water form acidic solution

Worksheet-3

1. **Correct Answer: C) To attain nearest noble gas configuration**

   Explanation: Atoms form bonds to achieve stability, usually by attaining the electronic configuration of the nearest noble gas. This also lowers their energy.

2. **Correct Answer: B) Ionic bond**

   Explanation: Ionic bonds (or electrovalent bonds) are formed when electrons are transferred from a metal (forms cation) to a non-metal (forms anion), held together by electrostatic attraction.

3. **Correct Answer: B) Mg and Cl**

   Explanation: Mg is divalent (+2) and Cl is monovalent (−1). So, $MgCl_2$ is formed to balance charges.

4. **Correct Answer: D) Both A and C**

   Explanation: Larger atoms lose electrons more easily (size effect). Cations with smaller positive charges are easier to form. Low ionization potential also favors cation formation.

5. **Correct Answer: B) Ionic charge increases**

   Explanation: Lattice energy depends on Coulomb's law. Higher charges lead to stronger attraction. Smaller ions also increase lattice energy because distance between ions decreases.

6. **Correct Answer: B) 6**

   Explanation: NaCl forms a face-centered cubic structure where each Na^+ is surrounded by 6 Cl^- ions and vice versa.

7. **Correct Answer: B) Mg**

   Explanation: Mg is in Group IIA. It has 2 valence electrons and loses both to achieve the nearest noble gas configuration, forming Mg^{2+}.

8. **Correct Answer: D) MX3**

   Explanation: Group IIIA metals (e.g., Al^{3+}) combine with Group VIA non-metals (e.g., O^{2-}). To balance charges:

   - 2 Al^{3+} ions + 3 O^{2-} ions → Al2O3.
   - So general formula = M2X3 or MX3 depending on notation.

9. **Correct Answer: C) Strong electrostatic attraction between ions**

   Explanation: Ionic solids have a 3D lattice with strong Coulombic forces between cations and anions, requiring high energy to break.

10. **Correct Answer: C) MgO**

    Explanation: Lattice energy increases with higher charges and smaller ionic radii. Mg^{2+} and O^{2-} both have high charges → strong attraction → highest lattice enthalpy.

Worksheet-4

1. **Correct Answer: B) ores**
   Explanation: Ores are minerals from which metals can be extracted economically.
2. **2. Correct Answer: B) refining**
   Explanation: Refining removes impurities from the crude metal to obtain pure metal.
3. **Correct Answer: C) aluminothermic**
   Explanation: In the thermite process, aluminium is used to reduce metal oxides.
4. **Correct Answer: C) calcination**
   Explanation: Calcination involves heating carbonate ores in the absence of air to form oxides.
5. **Correct Answer: D) (i) and (ii)**

   Explanation: Sodium and calcium are highly reactive metals and are extracted by electrolysis of their molten chlorides.

6. **Correct Answer: C) Zinc**

   Explanation: Zinc forms a protective layer and prevents iron from rusting.

7. **Correct Answer: B) $CuCO_3$**

   Explanation: The green coating is basic copper carbonate, formed by reaction with air, moisture, and carbon dioxide.

8. **Correct Answer: A) (i) and (ii)**

   Explanation: Gold and copper are commonly purified by electrolytic refining.

9. **Correct Answer: B) chemical; bauxite**

   Explanation: Bayer's process uses chemical affinity to separate bauxite from impurities.

10. **Correct Answer: D) Covering with glass**

    Explanation: Painting, anodizing, and alloying help prevent corrosion. Covering with glass is not a practical corrosion-prevention method for metals.

Competitive corner

SINGLE CORRECT OPTION TYPE

1. **Correct Answer: B) Cations**

   Explanation:
   Metals lose electrons from their outer (valence) shell. Losing negatively charged electrons makes them positively charged ions, called cations.

2. **Correct Answer: B) Negative**
   Explanation:
   Non-metals gain electrons to complete their valence shell, so they show negative valency.

3. **Correct Answer: C) Conc. HCl : Conc. HNO_3 = 3 : 1**
   Explanation:
   Aqua regia is a mixture of 3 parts concentrated hydrochloric acid (HCl) and 1 part concentrated nitric acid (HNO_3). It is strong enough to dissolve noble metals like gold.

4. **Correct Answer: B) (ii) and (iii)**
   Explanation:
   - KCl → ionic
   - NaCl → ionic
   - HCl → covalent (in pure form)
   - CCl_4 → covalent

5. **Correct Answer: B) Electrical conductivity in solid state**
   Explanation:
   Ionic compounds:
   - Conduct electricity when molten or dissolved in water
   - Do not conduct electricity in solid state because ions are fixed in position

6. **Correct Correct Answer: C) (ii) and (iv)**

   Explanation:
   Metals that are less reactive can exist freely in nature (native state):
   - Gold (Au)
   - Silver (Ag)
   - Zinc (Zn) is reactive, so it occurs as compounds
   - Copper (Cu) is usually found as compounds at this level

7. **Correct Answer: B**

   Explanation:
   In electrolytic refining of copper:
   - Impure copper is made the anode (+)
   - Pure copper is the cathode (–)
   - Electrolyte is acidified $CuSO_4$ solution
   - At the anode, $Cu \rightarrow Cu^{2+} + 2e^-$ (copper dissolves)
   - Cu^{2+} ions move to the cathode and get deposited as pure copper
   - Insoluble impurities fall down as anode mud

8. **Correct Answer: D) $CuSO_4$ + Fe**

   Explanation:
   A reaction occurs only if the metal is more reactive than the metal in the salt.
   Reactivity series:
   Mg > Zn > Fe > Pb > Cu
   - Fe can displace Cu from $CuSO_4$
   - $Fe + CuSO_4 \rightarrow FeSO_4 + Cu$

9. **Correct Correct Answer: A) Deposited on cathode**

   Explanation:
   In electrolytic refining:
   - Metal ions gain electrons at the cathode
   - Hence, pure zinc gets deposited on the cathode
   - Impurities either remain in solution or form anode mud

10. **Correct Answer: C) A homogeneous mixture**

    Explanation:
    An alloy is a uniform mixture of two or more metals (or a metal and a non-metal), e.g.:
    - Brass (Cu + Zn)
    - Steel (Fe + C)

    It shows uniform composition throughout, hence homogeneous.

11. **Correct Answer: C) Ag_2S**

    Explanation:
    Silver reacts with hydrogen sulfide (H_2S) present in air:
    - $2Ag + H_2S \rightarrow Ag_2S + H_2$
      Ag_2S (silver sulfide) is black in color
    - This causes the tarnishing of silver

12. **Correct Correct Answer: C) decreasing order of reactivity**

    Explanation:
    The activity (reactivity) series lists metals from most reactive at the top to least reactive at the bottom (e.g., K, Na, Ca … Cu, Ag, Au).
    Hence, metals are arranged in decreasing order of reactivity.

13. **Correct Correct Answer: A) Oxygen**

    Explanation:
    Oxygen makes up about 46% of the earth's crust by mass, mainly in the form of oxides and silicates.
    Therefore, oxygen is the most abundant non-metal.

14. **Correct Answer: A) iron forms ferrous sulphate solution**

    Explanation:
    Iron is more reactive than copper, so it displaces copper from copper sulphate:
    - $Fe + CuSO_4 \rightarrow FeSO_4 + Cu$
      $FeSO_4$ (ferrous sulphate) is light green
    - Copper is deposited as a reddish-brown solid

15. **Correct Correct Answer: A) $CuCO_3 \cdot Cu(OH)_2$**

Explanation:
Copper reacts slowly with moist air (oxygen + carbon dioxide + water) to form basic copper carbonate:

$$2Cu + H_2O + CO_2 + O_2 \rightarrow CuCO_3 \cdot Cu(OH)_2$$

This compound forms the green protective layer (patina) on copper.

ONE OR MORE THAN ONE CORRECT OPTION TYPE

16. **Correct Correct Answer: B) Leaching**

Explanation:
Leaching is a chemical method used to concentrate ores by dissolving the ore in a suitable chemical reagent and then extracting the metal from the solution.
Example: Bauxite is concentrated by leaching using NaOH.

17. **Correct Answer: A) C, B) CO,C)Al**

Explanation:

- Carbon (C) is used to reduce metal oxides (e.g., ZnO + C → Zn).
- Carbon monoxide (CO) reduces metal oxides (e.g., Fe_2O_3 + CO → Fe).
- Aluminium (Al) is used in aluminothermic reactions (e.g., thermite process).
  Since all are reducing agents, the correctCorrect Correct Answer :is All of these.

18. **Correct Answer: A) Platinum**

Explanation:
Copper sulphate solution reacts with more reactive metals like zinc, iron, or lead, causing displacement of copper.
Platinum **is** a noble metal and does not react**,** so it is safe.

19. **Correct Answers: A) Calcium and C) Potassium**

Explanation:

- Calcium (Ca) → Metal
- Potassium (K) → Metal
- Chlorine (Cl) → Non-metal
- Sulphur (S) → Non-metal

20. **Correct Answers: A) reducing agents and C) electropositive in nature**

Explanation:

- Metals lose electrons easily, so they act as reducing agents.
- Metals are electropositive, meaning they tend to form positive ions.

ASSERTION AND REASON TYPE

21. **Correct Correct Answer: (A)**

Explanation:
Metals are called electropositive because they lose electrons easily and form positive ions (cations)**.**
Thus, both the Assertion and Reason are correct, and the Reason correctly explains the Assertion.

22. **Correct Answer: (A)**

Explanation:
Potassium oxide (K_2O) reacts with water to form potassium hydroxide (KOH), which is a base**.**
Basic solutions turn red litmus blue, so the Reason correctly explains why potassium oxide is basic.

23. **Correct Answer: (A)**

Explanation:
Nitrogen has a very strong triple bond (N≡N) with high bond dissociation energy, making it chemically inert**.**
As a result, nitrogen does not readily form compounds, so nitrate ores are rare**.**
Hence, both statements are correct and the Reason explains the Assertion.

24. **Correct Answer: (A)**

Explanation:
Galvanisation involves coating iron with zinc. Zinc protects iron from rusting by forming a protective layer and acting as a sacrificial metal**.**
Therefore, both statements are correct and the Reason explains the Assertion.

MATRIX MATCHING TYPE

25. **Correct Answer: a) CaO → Basic oxide (R)**

Calcium oxide is a metal oxide. It reacts with acids to form salt and water, so it is basic in nature.

26. **b) Correct Answer: Al_2O_3 → Amphoteric oxide (P)**

Aluminium oxide reacts with both acids and bases, hence it is called amphoteric.

27. **Correct Answer: c) SO_2 → Acidic oxide (S)**

Sulphur dioxide is a non-metal oxide and reacts with water to form sulphurous acid (H_2SO_3), so it is acidic.

28. **Correct Answer:d) CO → Neutral oxide (Q)**

CO is held together by a strong triple covalent bond between carbon and oxygen, meaning electrons are shared rather than transferred, preventing ionization into acidic or basic species. so it is classified as a neutral oxide.

29. **Correct Answer: a) Iodine → Lustrous (R)**

Iodine is a non-metal, but it has a shiny (lustrous) appearance, which is an exception among non-metals.

30. **Correct Answer: b) Diamond → Hardest substance (S)**

Diamond is the hardest natural substance known and is an allotrope of carbon.

31. **Correct Answer: c) Mercury → Liquid metal (P)**

Mercury is the only metal that exists in liquid state at room temperature.

32. **Correct Answer: d) Bromine → Liquid non-metal (Q)**

Bromine is the only non-metal that exists in liquid state at room temperature.

COMPREHENSION TYPE

(I)

33. **Correct Answer: (C) Aluminium**

Explanation:
Aluminium is the most abundant metal found in the earth's crust (about 8%).
Iron is abundant, but aluminium is more abundant than iron in the crust.

34. **Correct Answer: (B) Sodium**

Explanation:
Sodium is a highly reactive alkali metal**.**
It reacts vigorously with cold water, producing sodium hydroxide and hydrogen gas.
Mercury, zinc, and tungsten do not react with cold water.

35. **Correct Correct Answer: (C) Magnesium**

Explanation:
Chlorophyll, the green pigment present in chloroplasts, contains magnesium at its center.
Magnesium plays a crucial role in photosynthes is**.**

(II)

36. **Correct Answer: D) Zinc plating**

Explanation:
Zinc plating (galvanisation) is the most durable protection for iron.
Zinc acts as a sacrificial metal—it corrodes first and protects iron even if the coating is damaged.

37. **Correct Answer: C) Connecting it with Mg block**

Explanation:
Magnesium is more reactive than iron.
When connected to iron, magnesium acts as a sacrificial anode and corrodes instead of iron, preventing rusting.
This method is called cathodic protection**.**

38. **Correct Answer: A) Making it cathode**

Explanation:
When iron is made the cathode, it does not lose electrons and hence does not undergo corrosion.
Putting iron in saline water actually increases rusting.

INTEGER TYPE

39. Correct Answer: 1

Explanation:
Only mercury (Hg) is a metal that exists in liquid state at room temperature.

40. Correct Answer: 2

Explanation:
Sodium (Na) and Potassium (K) are very soft metals and can be easily cut with a knife.

41. Correct Answer: 1

Explanation:
Iodine **is** the only common non-metal that is lustrous (shiny in appearance).

42. Correct Answer: 2

Explanation: Magnesium loses two electrons to form Mg^{2+}.

Previous years questions

1. Correct Answer: D

Nitric acid is a strong oxidising agent. It oxidises the hydrogen formed into water, so hydrogen gas is not evolved when most metals react with nitric acid.

2. Correct Correct Answer: D

- Aluminium forms Al_2O_3, which is amphoteric (reacts with acids and bases).
- Calcium forms CaO, which is a basic oxide.

3. Correct Answer: (B) I and IV only

I) The rate of evolution of H_2 gas bubbles must be the highest in test tube P and lowest in test tube S.

- P contains Ca (most reactive)
- S contains Pb (least reactive)

IV) The rate of evolution of H_2 gas bubbles is higher in test tube Q than in test tube R.

- Q → Al
- R → Fe
- Aluminium is more reactive than iron

4. Correct Answer: A

When Zinc reacts with sodium hydroxide, it forms sodium zincate and releases hydrogen gas:

$$Zn(s)+2NaOH(aq)\xrightarrow{\Delta}Na_2ZnO_2(aq)+H_2(g)\uparrow$$

5. Correct Answer: A

SO_3 (in Option B) forms H_2SO_4, which is a strong acid. CO (in Option C) is a neutral oxide and does not form an acid with water.

6. Correct Answer: B

Explanation: Metal P: Highest number of bubbles (Most reactive).
Metal Q: High number of bubbles, but fewer than P.
Metal R: Moderate number of bubbles.
Metal S: Very few bubbles (Slow reaction).
Metal T: No bubbles at all (No reaction).

7. Correct Answer: C.

Arranging these metals from the least reactive to the most reactive (using the < symbol), we get: Lead (Pb) < Zinc (Zn) < Aluminum (Al) < Calcium (Ca) < Sodium (Na)

8. Correct Answer: C

Froth floatation process, Conversion to metal oxide and reduction of metal oxide.

9. Correct Answer: (A) Al_2O_3.

Explanation: Aluminum oxide is a metallic oxide that is amphoteric, meaning it reacts with (and dissolves in) both acids and bases. CaO and Na_2O are purely basic, and CO_2 is a non-metal acidic oxide.

10. Correct Answer: A Refining, electrolytic refining, anode, cathode.

The process of removing impurities from impure metal to obtain pure metal is known as refining of metal.
The most widely used method for refining is electrolytic refining.
In this method, the impure metal is taken as anode.
The pure metal is taken as cathode.

3. CHEMICAL REACTIONS AND EQUATIONS

Solutions

Worksheet – 1

1. **Correct Answer: A) Created, destroyed**
   Explanation**:** This is the Law of Conservation of Mass. It states that matter cannot be created or destroyed in a closed system.
2. **Correct Answer: B) $3CuSO_4 + KI \rightarrow Cu_2I_2 + 2I_2 + 2K_2SO_4$**
   Explanation: Let's count the atoms on both sides for option B:
   - Copper (Cu): 3 on the left, 2 on the right. (Unbalanced)
   - Potassium (K): 1 on the left, 4 on the right. (Unbalanced)
   - Iodine (I): 1 on the left, 6 on the right (2+4). (Unbalanced)
   - Since the number of atoms doesn't match, it is not balanced.
3. **Correct Answer: C) $2FeS + 3O_2 \rightarrow 2FeO + 2SO_2$**
   Explanation: Let's verify the counts for Option C:
   - Fe: 2 on left, 2 on right. (Balanced)
   - S: 2 on left, 2 on right. (Balanced)
   - O: 6 on left (3 X2), 6 on right (2 + [2 X 2]). (Balanced)
4. **Correct Answer: D) Both A and B**
   Explanation:
   - In A: $(NH_4)PO_4$ $3NH_3 + H_2O + HPO_3$. Left has 1 Phosphorus; Right has 1. Left has 1 Nitrogen; Right has 3. Unbalanced.
   - In B: $N_2 + 3H_2 \longrightarrow 4NH_3$. Left has 2 Nitrogen; Right has 4. Left has 6 Hydrogen; Right has 12. Unbalanced.
5. **Correct Answer: A) O_2**
   Explanation: We balance the atoms we have:
   - Left side: 2 Potassium (K), 2 Nitrogen (N), 6 Oxygen (O).
   - Right side (so far): 2 Potassium (K), 2 Nitrogen (N), 4 Oxygen (O).
   - To balance the oxygen, X must contain the remaining 2 Oxygen atoms. Therefore, X = O_2.
6. **Correct Answer: A) n + m = p + q**
   Explanation: According to the Law of Conservation of Mass, the total mass of reactants = total mass of products.
   - Reactants mass: n + m
   - Products mass: p + q
   - Therefore, n + m = p + q
7. **Correct Answer: B) $Cu + 2AgNO_3 \rightarrow Cu(NO_3)_2 + 2Ag$**
   Explanation: Copper (Cu) has a valency of +2 when forming Copper(II) Nitrate, while the Nitrate ion (NO_3) has a valency of -1. Therefore, the formula for the product must be $Cu(NO_3)_2$
   - To balance the two NO_3 groups on the right, we need 2AgNO on the left.
   - To balance the 2Ag atoms now on the left, we need 2Ag on the right.
8. **Correct Answer: B)**
   Explanation: Identify Reactants and Products:
   Nitrogen (N_2) + Hydrogen (H_2) $\longrightarrow$Ammonia (NH_3).
   Count Nitrogen atoms: There are 2 Nitrogen atoms on the left (N_2). To match this, we need a coefficient of 2 for NH_3 on the right.
   - Result: $N_2 + H_2 \rightarrow 2NH_3$

   Count Hydrogen atoms: Now there are 2 X 3 = 6 Hydrogen atoms on the right. To get 6 Hydrogen atoms on the left, we need a coefficient of **3** for H2 (3 X 2 =6).
   - Result: $N_2 + 3H_2 \longrightarrow 2NH_3$

9. Correct correct: A

standard balanced equation does not tell us the speed (rate) of the reaction. It doesn't show how fast the reactants turn into products; that information belongs to the field of Chemical Kinetics.

10. Correct Answer: C)

Explanation:
A chemical equation is called balanced when the number of atoms of each element in the reactants is equal to the number of atoms of the same element in the products, in accordance with the law of conservation of mass

Worksheet-2

1. Correct Answer: B) Two moles of phosphine

Explanation: Chemical Equation: Ca_3P_2 + $6H_2O$ → $3Ca(OH)_2$ + $2PH_3$

Reasoning: Looking at the balanced equation, 1 mole of Ca_3P_2 reacts to produce 2 moles of phosphine (PH_3).

2. Correct Answer: D) 0.1

Explanation: Identify Limiting Reagent:

- For $BaCl_2$: 0.5 mol 3 = 0.166
- For Na_3PO_4: 0.2 mol 2 = 0.10 (Limiting Reagent)
- Calculation: According to the stoichiometry, 2 moles of Na_3PO_4 produce 1 mole of $Ba_3(PO_4)_2$

$$\text{Moles produced} = 0.2 \text{ mol } Na_3PO_4 \times \left(\frac{1 \text{ mol } Ba_3(PO_4)_2}{2 \text{ mol } Na_3PO_4}\right) = 0.1 \text{ mol}$$

3. Correct Answer: A) 2000 g

Molar Masses:

$$Octane(C_8H_{18}) \approx 114 \text{ g/mol}; Oxygen(O_2) \approx 32 \text{ g/mol}$$

$$\text{Moles of octane} = \frac{570 \text{ g}}{114 \text{ g/mol}} = 5 \text{ moles}$$

$$\text{Moles of } O_2 \text{ required} = 5 \text{ mol octane} \times \left(\frac{25 \text{ mol } O_2}{2 \text{ mol octane}}\right) = 62.5 \text{ moles}$$

$$\text{Mass of } O_2 = 62.5 \text{ mol} \times 32 \text{ g/mol} = 2000 \text{ g}$$

4. Correct Answer: B) 106.41 kg

$$Pure\ Reactant\ Mass: 200 \text{ kg} \times 0.95 = 190 \text{ kg of pure } CaCO_3.$$

$$Molar mass\ of\ CaCO_3 = 100 \text{ g/mol}; Molar mass\ of\ CaO = 56 \text{ g/mol}.$$

$$\text{Mass of } CaO = 190 \text{ kg } CaCO_3 \times \left(\frac{56 \text{ kg } CaO}{100 \text{ kg } CaCO_3}\right) = 106.4 \text{ kg}$$

5. Correct Answer: A) 22.12 g

$$HCl \approx 36.5 \text{ g/mol}; MgCl_2 \approx 95.3 \text{ g/mol}$$

$$\text{Moles of } HCl = \frac{17 \text{ g}}{36.5 \text{ g/mol}} \approx 0.466 \text{ moles}$$

$$\text{Moles of } MgCl_2 = 0.466 \text{ mol } HCl \times \left(\frac{1 \text{ mole } MgCl_2}{2 \text{ moles } HCl}\right) \approx 0.233 \text{ moles}$$

$$\text{Mass of } MgCl_2 = 0.233 \text{ mol} \times 95.3 \text{ g/mol} \approx 22.2 \text{ g}$$

6. Correct Answer: A) 42 g

$$\text{Molar mass of } H_2O = 18\ gmol$$

$$\text{Moles of steam} = \frac{18}{18} = 1 \text{ mol}$$

$$Molar Masses: Fe = 56 \text{ g/mol}; H_2O = 18 \text{ g/mol}$$

Stoichiometry: According to the equation, 4 moles of H_2O react with 3 moles of Fe

$$Moles\ of\ F\ rerequired: 1 \text{ mole } H_2O \times \left(\frac{3 \text{ moles } Fe}{4 \text{ moles } H_2O}\right) = 0.75 \text{ moles}$$

$$Mass\ of\ Fe: 0.75 \text{ moles} \times 56 \text{ g/mol} = 42 \text{ g}$$

7. Correct Answer: D) 64 g

Molar mass of H_2O = 18 g

So, 18 g H_2O contains 16 g of oxygen

$$72 \text{ g } H_2O = \frac{72}{18} = 4 \text{ moles}$$

Oxygen obtained:

$$4 \times 16 = 64 \text{ g}$$

8. Correct Answer: X grams of $CaCO_3$ completely burned, residue = 28 g

On heating:

$$CaCO_3 \rightarrow CaO + CO_2$$

Molar masses:

$CaCO_3$ = 100 g; CaO = 56 g

So,100 g $CaCO_3$ → 56 g CaO

Residue is CaO = 28 g

$$CaCO_3 \text{ used} = \frac{100}{56} \times 28 = 50 \text{ g}$$

9. **Correct Answer: Reaction:**

$$4Al + 3O_2 \rightarrow 2Al_2O_3$$

Molar masses:

Al = 27 g ;O = 13.5 g

From the balanced equation, 4 moles of Al react with 3 moles ofO_2 . The molar ratio of O_2 to Al is 3:4

$n_{O_2} = n_{Al} \times \frac{3mol\ O_2}{4mol\ Al} = 0.5mol \times \frac{3}{4} = 0.375mol$

nO_2=nAl×3mol O_2 4molAl =0.5mol×34=0.375mol

Using the molar mass of oxygen gas (MO^2=2×16=32g/mol), the required mass of oxygen ($_2mO2$) is:

$m_{O_2} = n_{O_2} \times M_{O_2} = 0.375mol \times 32g/mol = 12g$

mO_2=nO_2×MO_2=0.375mol×32g/mol=12g

So, Correct Answer: 12 g (Option C)

10. **Correct Answer: Given:**

CO_2 formed = 4.4 g

Molar mass of CO_2 = 44 g

Carbon in 44 g CO_2 = 12 g

Carbon in 4.4 g CO_2:

$$\frac{12}{44} \times 4.4 = 1.2\ g$$

Mass of compound = 1.6 g

Percentage of carbon:

$$\frac{1.2}{1.6} \times 100 = 75\%$$

So, Correct Answer: 75% (Option D)

Worksheet-3

1. **Correct Answer: A) Chemical combination**

When two or more elements or compounds react to form one single product, the reaction is called a chemical combination (or synthesis) reaction.

2. **Correct Answer:** A) Combination reaction

Explanation:

In a combination reaction, two or more substances combine to form a single product.

3. **Correct Answer:** B) Chemical decomposition

Explanation:

In a decomposition reaction, one compound breaks down into two or more simpler substances.

4. **Correct Answer:** B) AgCl – white precipitate

Explanation:

This is a double displacement reaction:

- Silver nitrate reacts with sodium chloride
- Silver chloride (AgCl) is formed as a white precipitate
- Sodium nitrate ($NaNO_3$) remains in solution.

5. **Correct Answer: B) Ca + Cl_2 → $CaCl_2$**

Explanation:

- A single displacement reaction involves one element displacing another from a compound.
- In option B, calcium and chlorine combine directly to form $CaCl_2$.
- No element is displaced.

6. **Correct Answer: B) Single displacement reaction**

Explanation:

- Zinc is more reactive than hydrogen
- Zinc displaces hydrogen from sulphuric acid
- Hydrogen gas is released

7. **Correct Answer: B) Na + NaCl → NaCl + Na**

Explanation:

- A metal cannot displace itself from its compound
- Sodium cannot displace sodium from sodium chloride

8. **Correct Answer: C) Cu + 2$AgNO_3$ → $Cu(NO_3)_2$ + 2Ag**

Explanation:Copper is more reactive than silver

Copper displaces silver from silver nitrate solution, This follows the reactivity series

9. **Correct Answer: D) Pt (Platinum)**

Explanation: Platinum is at the bottom of the reactivity series

It is very unreactive, Therefore, it cannot displace other elements from their compounds

10. **Correct Answer: B)** ***Correct***
In a displacement reaction, one element displaces another element from its compound.
Example:
$Zn + CuSO_4 \rightarrow ZnSO_4 + Cu$

Worksheet-4

1. **Correct Answer: A) $2H_2O \rightarrow 2H_2 + O_2(g)$**
Explanation:
An electrochemical reaction involves the use of electric current to bring about a chemical change (electrolysis).
2. **Correct Answer: C) Photochemical reactions**
Explanation:
   - Photochemical reactions require light energy to proceed.
   Example: Photosynthesis.
3. **Correct Answer: B) CuO**
Explanation:
   - Oxidising agent is the substance that gains oxygen or loses hydrogen.
   - CuO gives oxygen to hydrogen and gets reduced to Cu.
   - Hence, CuO is the oxidising agent.
4. **Correct Answer: D) $CaCO_3 \rightarrow CaO + CO_2$**
Explanation:
   - Endothermic reactions absorb heat.
   - Decomposition of calcium carbonate requires heat.
   - Therefore, it is an endothermic reaction.
5. **Correct Answer: B) Zinc acts as a sacrificial metal**
Explanation:
   - Zinc is more reactive than iron.
   - Even if scratched, zinc corrodes first and protects iron.
   - This is called sacrificial protection.
6. **Correct Answer: C) Preventing oxidation of fats**
Explanation:
   - Nitrogen is an inert gas.
   - It prevents fats and oils from reacting with oxygen.
   - This stops rancidity and keeps food fresh.
7. **Correct Answer: C) Chips developing bad smell after long storage**
Explanation:
   - Rancidity is caused by oxidation of fats and oils.
   - It results in bad smell and taste.
   - Chips contain oils, so they become rancid over time.
8. **Correct Answer: C. Precipitation reaction**
   - Explanation:
   A precipitation reaction forms an insoluble solid (precipitate) when two aqueous solutions react.
9. **Correct Answer:B. $NaCl(aq) + AgNO_3(aq) \rightarrow AgCl(s) + NaNO_3(aq)$**
Explanation:
   - AgCl (silver chloride) is insoluble in water and forms a solid precipitate.
10. **Correct Answer:B. $HCl + NaOH \rightarrow NaCl + H_2O$**
Explanation:
   - A neutralisation reaction occurs when an acid reacts with a base to form salt and water.

Competitive Corner

SINGLE CORRECT OPTION TYPE

1. **Correct Answer: A) $N_2 + 3H_2$ $2NH_3$**
   - Left: 2N, 6H | Right: 2N, 6H. (Balanced)
2. **Correct Answer: B**

Copper (Cu) displaces Silver (Ag) from silver nitrate. Copper forms Cu^{2+} ions, so the formula for copper(II) nitrate is $Cu(NO_3)_2$.

   - Equation: $Cu + 2AgNO_3 \rightarrow Cu(NO_3)_2 + 2Ag$
   - Check: 1Cu, 2Ag, 2N, 6O on both sides.

3. Correct Answer: D

The decomposition reaction is: $2H_2O \rightarrow 2H_2 + O_2$.

1. Molar Mass of H_2O: 2(1) + 16 = 18 gmol.
2. Moles of H_2O: $\frac{72g}{18g/mol}$ = 4 moles.
3. Mole Ratio: According to the equation, 2 moles of H_2O produce 1mole of O_2
4. So, 4 moles of H_2O will produce 2 moles of O_2.
5. Mass of O_2: 2 moles X 32 gmol = 64g.

4. Correct Answer: D

- Molar Mass of CaCO3**:** 40 + 12 + 3(16) = 100 gmol.
- Molar Mass of CaO**:** 40 + 16 = 56 gmol.
- Stoichiometry: 100 g of CaCO3 produces 56 g of CaO.
- Calculation: If 56 g of residue comes from 100g of CaCO3, then 28 g (which is half) must come from half the amount of CaCO3.
- X = $\frac{100X28}{56}$ = 50g.

5. Correct Answer: C

The reaction is: $4Al + 3O2 \rightarrow 2Al_2O_3$.

Atomic Mass Al: 27 gmol.

Moles of Al: $\frac{13.5g}{27g/mol}$ = 0.5 moles.

Mole Ratio: 4 moles Al react with 3 moles O_2.

- 0.5 moles Al react with $\frac{3}{4}$ X 0.5 = 0.375 moles O_2.

Mass of O_2**:** 0.375 moles X 32 gmol = 12 g.

6. All carbon in CO2 comes from the organic compound.

Mass of Carbon in 4.4 g CO2:

- Molar mass of CO2 = 44 g\mol. Carbon part is 12 g.
- Mass of C = $\frac{12}{44}$ X 4.4 g = 1.2 g.

Percentage of Carbon:

- %C = $\frac{Mass\,of\,carbon}{Total\,mass\,of\,compound}$ X 100
- %C = $\frac{1.2}{1.6}$ X 100 = 0.75 X 100 = 75%.

Answer: D

7. Correct Answer is C) Chemical synthesis.

In chemistry, a **synthesis reaction** (also known as a direct combination reaction) occurs when two or more simple substances—either elements or simpler compounds—combine to form a single, more complex product.

8. Correct Answer: A) Chemical combination

Explanation: A combination reaction occurs when two or more reactants combine to form a single product. Here, calcium oxide (quicklime) reacts with water to form a single substance, calcium hydroxide (slaked lime).

9. Correct Answer: A) Na, Fe, Cu, Ag

Explanation**:** Based on the metal reactivity series, Sodium (Na) is highly reactive (alkali metal), followed by Iron (Fe). Copper (Cu) and Silver (Ag) are noble metals found near the bottom of the series because they are very unreactive.

10. Correct Answer: C. Adding antioxidants

Explanation: Rancidity is caused by the oxidation of fats and oils. Antioxidants are substances that neutralize free radicals and prevent oxygen from reacting with the food, thereby preserving it.

11. **Correct Answer: C. To reduce contact with oxygen**
Explanation**:** Oxygen in the air reacts with oils (oxidation), leading to a foul smell and taste (rancidity). Airtight containers limit the supply of oxygen, slowing this process down.
12. **Correct Answer: C. ↓**
Explanation**:** ↓ (Downward arrow) indicates a solid precipitate settling out of a solution.
    - ↑ (Upward arrow) indicates the evolution of a gas.
    - -→ indicates the direction of the reaction.
    - ⇌ indicates a reversible reaction.
13. **Correct Answer: B. Exothermic**
Explanation**:** A neutralization reaction occurs between an acid and a base to produce salt and water. This process almost always releases energy in the form of heat, making it exothermic.

ONE OR MORE THAN ONE CORRECT OPTION TYPE

14. **Correct Answer: A, B, and C**
Explanation: Corrosion is the gradual destruction of metals by chemical or electrochemical reactions with their environment (moisture and air).
Rusting of iron (A): Forms hydrated iron(III) oxide (reddish-brown).
Green coating on copper (B): Forms basic copper carbonate due to moist CO_2.
Tarnishing of silver (C): Forms silver sulfide (Ag_2S) due to sulfur in the air, turning it black.
15. **Correct Answer: A, B, and C**
Explanation: Rancidity is the process of oxidation of fats and oils in food. This chemical change results in:
    - An unpleasant change in smell (C).
    - A foul change in taste (B).
16. **Correct Answer: A, B, and C**
Explanation**:** A precipitate is an insoluble solid that emerges from a liquid solution during a chemical reaction.
    - AgCl (A): Silver Chloride is a classic white precipitate.
    - $BaSO_4$ (B): Barium Sulfate is a heavy white precipitate.
    - AgBr (C): Silver Bromide is a pale yellow precipitate (used in photography)
17. **Correct Answer: B and D**
Explanation: A neutralization reaction occurs between an Acid and a Base.
    - Heat energy is released (B**):** These reactions are almost always exothermic.
    - Salt formation occurs (D): The general equation is: Acid + Base -→ Salt + Water.
18. **Correct Answer: A, B, and D**
Explanation: Key indicators of a chemical reaction include a permanent change in color (A), the formation of a precipitate (B), and the evolution of a gas (D).
19. **Correct Answer: A, B, and D**
Explanation: A decomposition reaction occurs when one reactant breaks down into two or more simpler products.
    - **A:** Lead nitrate breaks down into lead oxide, nitrogen dioxide, and oxygen.
    - **B:** Calcium carbonate breaks down into calcium oxide and carbon dioxide.
    - **D:** Silver chloride decomposes into silver and chlorine gas when exposed to light.
20. **Correct Answer: A, B, and D**
Explanation**:** In double displacement reactions, two compounds exchange ions to form two new products.
    - **A:** Sodium sulfate and Barium chloride swap ions to form a Barium sulfate precipitate.
    - **B:** Silver nitrate and Sodium chloride swap ions to form Silver chloride.

- **D:** Hydrochloric acid and Sodium hydroxide swap ions to form salt and water (neutralization).

21. **Correct Answer: A,B,Cand D**
    - **$2Cu + O_2 \rightarrow 2CuO$**: In this reaction, **copper (Cu)**'s oxidation state increases from 0 to +2 (it gains oxygen).
    - **$Zn + CuSO_4 \rightarrow ZnSO_4 + Cu$**: Here, **zinc (Zn)**'s oxidation state increases from 0 to +2 (it loses electrons), while copper's oxidation state decreases from +2 to 0.
    - **$Fe_2O_3 + 3CO \rightarrow 2Fe + 3CO_2$**: In this reaction (used in iron extraction), **carbon monoxide (CO)** is oxidized to carbon dioxide (CO_2), as carbon's oxidation state increases from +2 to +4, and oxygen is gained by the carbon species.
    - **$C + O_2 \rightarrow CO_2$**: Here, **carbon (C)** is oxidized from an oxidation state of 0 to +4 (it gains oxygen).

ASSERTION AND REASON TYPE

22. **Correct Answer: A. Both A and R are true and R is the correct explanation of A**
    Explanation: * Assertion **(A):** Decomposition reactions involve a single reactant breaking down into simpler products. This process is not spontaneous and requires an external source of energy like heat, light, or electricity.
    Reason (R): Chemical bonds hold atoms together in a molecule. To break these bonds and allow the substance to decompose, energy must be absorbed from the surroundings. Since the reason explains *why* the energy is required, it is the correct explanation.
23. **Correct Answer: C. A is true but R is false**
    Explanation:
    Assertion (A): Photosynthesis is indeed an endothermic reaction because plants absorb solar energy (sunlight) to convert carbon dioxide and water into glucose.
    Reason (R): This statement is false. Energy is absorbed, not released, during photosynthesis. Reactions that release energy are called exothermic reactions.
24. **Correct Answer: A. Both A and R are true and R is the correct explanation of A**
    Explanation:
    - Assertion (A): In a displacement reaction, a more reactive element displaces a less reactive element from its compound.
    - Reason (R): During this process, the more reactive metal loses electrons (oxidation), and the less reactive metal ion gains electrons (reduction). Since oxidation and reduction happen at the same time, it is by definition a redox reaction.
25. **Correct Answer: A. Both A and R are true and R is the correct explanation of A**
    Explanation:
    - Assertion (A): The reaction H2 + Cl2 $\xrightarrow{sunlight}$ 2HCl is a valid chemical process where hydrogen chloride gas is formed.
    - Reason (R): This is a combination reaction because two different elements (reactants) combine to form a single new product. The reason correctly identifies the type of reaction described in the assertion.

MATRIX MATCHING TYPE

26. **a) Combination Reaction**
    - Reaction**:** Calcium oxide (quicklime) reacts with water to form calcium hydroxide (slaked lime).
    - Explanation: A combination reaction occurs when two or more reactants combine to form a single product. In this case, two compounds (CaO and H2O) merge into one.

b) Displacement Reaction

- Reaction**:** Zinc metal reacts with copper sulfate solution to produce zinc sulfate and solid copper.
- Explanation: A displacement reaction (or single displacement) happens when a more reactive element displaces a less reactive element from its compound. Zinc is more reactive than copper, so it "kicks out" the copper from the sulfate.

c) Double Displacement Reaction

- Reaction: Silver nitrate and sodium chloride exchange ions to form silver chloride and sodium nitrate.
- Explanation: In double displacement reactions, two compounds exchange ions to form two new products. Often, one of the products is an insoluble precipitate, like the AgCl seen here.

d) Decomposition Reaction

- Reaction**:** Calcium carbonate (limestone) breaks down into calcium oxide and carbon dioxide when heated.
- Explanation: A decomposition reaction is the opposite of combination; it occurs when a single reactant breaks down into two or more simpler products. Because this specific reaction requires heat, it is also known as thermal decomposition.

27. a) Rusting –→ Oxidation & Corrosion

- **Explanation:** Rusting is a specific type of **corrosion** that occurs when iron (Fe) reacts with oxygen and moisture in the air.
- **Process:** Chemically, it is an **oxidation** reaction because iron atoms lose electrons to oxygen to form hydrated iron(III) oxide (Fe2O3} . ^{x}H2O).

b) Photosynthesis –→ Endothermic Reaction

- **Explanation:** Photosynthesis is a process used by plants to convert light energy into chemical energy.
- **Process:** It is considered an **endothermic reaction** because it requires a continuous input of energy (sunlight) to break the bonds of carbon dioxide and water to produce glucose.

c) Respiration –→ Exothermic Reaction

- **Explanation:** Respiration is the process of breaking down glucose in cells to provide energy for life processes.
- Process: It is an exothermic reaction because energy is released in the form of heat and ATP during the breakdown of chemical bonds.

d) Black coating on silver –→ Oxidation & Corrosion

- **Explanation:** Silver develops a black coating when it reacts with hydrogen sulfide gas in the air, forming silver sulfide (Ag_2S).
- **Process:** This is a form of **corrosion** (often called tarnishing). It is also an **oxidation** reaction because the silver metal (Ag) loses electrons during the formation of the sulfide compound.

COMPREHENSION TYPE

(I)

28. Correct Answer: (B) 4Na + O2 –→ 2Na2O

Reasoning**:** A combination reaction involves two or more reactants forming a single product. The question specifically asks for the combination of two elements.

29. **Correct Answer: (D) simple displacement reaction**
Reasoning**:** In this reaction, Aluminum (Al) is more reactive than Iron (Fe). It "displaces" or kicks out the iron from its oxide to form Aluminum Oxide. Since only one element is being displaced from a compound, it is a simple displacement reaction (also known as a thermite reaction).

30. **Correct Answer: (C) (ii) and (v)**
Reasoning: * (ii) Displacement reaction**:** Magnesium (Mg) displaces Copper (Cu) from its oxide because Magnesium is more reactive.
(v) Redox reaction**:** This is also a reduction-oxidation (redox) reaction. Magnesium is oxidized (gains oxygen to become MgO), and Copper Oxide is reduced (loses oxygen to become Cu). (II)

31. **Correct Answer: C. Copper**
Explanation:
    - Copper(II) sulfate (CuSO4) is famously known as "blue vitriol" because its aqueous solution is bright blue.
    - This blue color is specifically due to the presence of hydrated copper ions (Cu^{2+}) in the solution.
    - In contrast, Zinc and Magnesium sulfates form colorless solutions, while Iron(II) sulfate ($FeSO_4$) solutions are typically pale green

32. **Correct Answer: B. $BaSO_4$**
Explanation:
    - When barium chloride ($BaCl_2$) reacts with any soluble sulfate salt (like $CuSO_4$ or Na_2SO_4), a chemical reaction occurs where the barium ions (Ba^{2+}) and sulfate ions (SO_4^{2-}) combine.
    - This forms Barium Sulfate ($BaSO_4$), which is a highly insoluble solid that appears as a white precipitate in the test tube.
    - The reaction for salt X ($CuSO_4$) is:

      $BaCl_2$ (aq) + $CuSO_4$ (aq) + $BaSO_4$ (s) + $CuCl_2$ (aq)
      The $CuCl_2$ formed is the salt that turns the solution green.

33. **Correct Answer: D. Double displacement reaction**
Explanation:
    - In this reaction, the two reacting compounds "exchange" their partners: the Barium (Ba) swaps places with Copper (X), and the Chloride (Cl) swaps with Sulfate (SO_4).
    - Because two different atoms or groups of atoms are being displaced and exchanged simultaneously, it is defined as a double displacement reaction.
    - It is also specifically categorized as a precipitation reaction because one of the products formed is an insoluble solid.

INTEGER TYPE

34. **Correct Answer: The number of oxygen atoms present in $2CO_2$ is 4.**
Explanation:
    - The Molecule**:** One molecule of carbon dioxide (CO_2) contains 1 Carbon (C) atom and 2 Oxygen (O) atoms (indicated by the subscript $_2$).
    - The Coefficient**:** The large number in front (2) is the coefficient, which tells us there are two molecules of CO_2.
    - Calculation: To find the total atoms, you multiply the coefficient by the subscript: $2 \times 2 = 4$ oxygen atoms.

35. **Correct Answer: The number of products formed in this reaction is 2.**
Explanation:
    - Reactants vs. Products**:** In a chemical equation, substances on the left of the arrow are reactants, and substances on the right are products.

- Identification**:** Here, Calcium Carbonate ($CaCO_3$) is the single reactant. It breaks down into two distinct substances: Calcium Oxide (CaO) and Carbon Dioxide (CO_2).
- Type**:** This is specifically called a Thermal Decomposition reaction because one substance splits into two or more products when heated.

36. Correct Answer: This is a Single Displacement (or Replacement) reaction.

Explanation:

- Mechanism**:** In this reaction, Zinc (Zn) is more reactive than Copper (Cu). Because of its higher reactivity, Zinc "kicks out" or displaces Copper from its compound, Copper Sulfate ($CuSO_4$).
- Result: Zinc combines with the sulfate ion to form Zinc Sulfate ($ZnSO_4$), leaving Copper as a pure metal.
- Redox Note: It is also a Redox (Reduction-Oxidation) reaction, as Zinc loses electrons (oxidation) and Copper ions gain electrons (reduction).

37. Correct Answer: The number of different types of chemical reactions involved in the rusting of iron is generally considered 2 (Oxidation and Combination/Hydration).

Explanation:

Rusting is a complex electrochemical process, but it can be summarized in these chemical stages:

Oxidation (Redox): Iron (Fe) reacts with oxygen in the presence of water. Iron atoms lose electrons to form iron ions.

Hydration (Combination)**:** The iron oxide then combines with a variable number of water molecules (xH_2O) to form the flaky brown substance we call rust (hydrated iron(III) oxide).

PREVIOUS YEAR QUESTIONS

1. Correct Answer: Step 1: Identify Oxidation and Reduction

- Reduction: Manganese (Mn) goes from an oxidation state of +4 in MnO_2 to 0 in Mn. Thus, MnO_2 is the substance being reduced.
- Oxidation**:** Aluminum (Al) goes from an oxidation state of 0 to +3 in Al_2O_3. Thus, Al is the substance being oxidized.

Correct Answer: Step 2: Balance the Equation

We can balance this using the hit-and-trial method or by balancing atoms:

1. Look at Oxygen: There are 2 on the left and 3 on the right. To balance, we use a common multiple of 6.
2. Place a 3 before MnO_2 and a 2 before Al_2O_3.
3. Now balance the metals:
   - 3 Mn on the left → place a 3 before Mn.
   - 4 Al on the right (2 X 2) place a 4 before Al.

Balanced Equation:

$3MnO_2 + 4Al \rightarrow 3Mn + 2Al_2O_3$

Step 3: Determine the Ratio

- Coefficient of substance reduced (MnO_2) = 3
- Coefficient of substance oxidized (Al) = 4

Ratio = 3:4

So, Correct Answer is A

2. Correct Answer: C

Correct**:** Sodium bicarbonate is indeed used as an antacid (to neutralize stomach acid) and is the primary component in soda-acid fire extinguishers.

3. Correct Answer: B

Analysis: When an electric current is passed through water, it decomposes into its constituent elements: hydrogen gas and oxygen gas.

Chemical Equation**:** $2H_2O(l) \xrightarrow{electricity} 2H_2(g) + O_2(g)$

Conclusion**:** Since a single compound (water) splits into two simpler substances, it is a decomposition reaction.

4. **Correct Answer: A**
   Analysis**:** A classic double displacement reaction that produces a characteristic **y**ellow precipitate involves Lead (II) nitrate and Potassium iodide.
5. **Correct Answer: C**
   H_2 is being oxidized, not reduced, because it gains oxygen to form H_2O.
6. **Correct Answer: D) 2 and 4**
   - Pb: 2 on left, 2 on right.
   - N: 4 on left, 4 on right.
   - O: 12 on left (2 X 3 X 2), 12 on right (2 + 8 + 2).
7. **Correct Answer: A) (ii), (iii) and (iv) only**
   (ii) Decomposition of vegetable matter into compost – Endothermic reaction: Incorrect**.** This is a biological decomposition process that releases energy (heat), making it exothermic.
   (iii) Reaction of zinc with copper sulphate – Decomposition reaction: Incorrect. This is a Single Displacement reaction ($Zn + CuSO^4 \rightarrow ZnSO^4 + Cu$).
   (iv) Reaction of barium chloride with sodium sulphate – Single displacement reaction: Incorrect. This is a Double Displacement (Precipitation) reaction ($BaCl2 + Na2SO^4 \rightarrow BaSO^4 + 2NaCl$).
8. **Correct Answer: C) 2, 2 and 3**
   - Place a 2 before KClO3 (2 X 3 = 6 Oxygens).
   - Place a 3 before O2 (3 X 2 = 6 Oxygens).

   2KClO3 → 2KCl + 3O2 ;x=2, y=2, and z=3
9. **Correct Answer:**
   $NaOH_{(aq)} + HCl_{(aq)} \rightarrow NaCl_{(aq)} + H_2O_{(l)}$
   Sodium (Na): Remains +1 on both sides.
   - Oxygen (O): Remains -2 on both sides.
   - Hydrogen (H): Remains +1 on both sides.
   - Chlorine (Cl): Remains -1 on both sides.
   - Verdict: This is an acid-base neutralization reaction. There is no transfer of electrons or change in oxidation states. This is NOT a redox reaction.
10. **Correct Answer: C) Oxidising agent, Oxidation, reduction, Reducing agent**
    - Zinc Oxide (ZnO) to Zinc (Zn): ZnO loses oxygen to become Zn. The removal of oxygen is reduction. Because ZnO provides oxygen to the other reactant, it acts as the oxidising agent.
    - Carbon (C) to Carbon Monoxide (CO): Carbon gains oxygen to become CO. The gain of oxygen is oxidation. Because Carbon removes oxygen from ZnO, it acts as the reducing agent.

4. Acids, Bases and Salts

Solutions

Worksheet – 1

1. **Correct Answer: C**
   The word "acid" originates from the Latin term acidus, which specifically translates to sour.
2. **Correct Answer: D**
   Citric acid is the primary organic acid in citrus fruits such as lemons.
3. **Correct Answer: D**
   HCl is a mineral acid because it is derived from inorganic mineral sources, unlike the other options which are organic carboxylic acids.
4. **Correct Answer: B**
   Blue litmus paper changes to red color in the presence of an acid.
5. **Correct Answer: D**
   Phenolphthalein is a pH indicator that remains colorless in acidic or neutral media but turns pink in basic solutions
6. **Correct Answer: D**
   $[H_3O^+] = 10^{-pH}$; $\text{Ratio} = \frac{10^{-2}}{10^{-4}} = 100$
7. **Correct Answer: C**
   Acids derived from biological sources (plants and animals) are classified as organic acids, typically containing the carboxyl group ($-COOH$).
8. **Correct Answer: C**
   Sodium hydroxide (NaOH) is a strong Arrhenius base in aqueous solution due to the hydrolysis of the carbonate ion, which produces hydroxide ions.
9. **Correct Answer: B**
   $pH = -\log[H^+]$; $pH = -\log(1) = 0$
10. **Correct Answer: D**
    $[H^+] = 10^{-pH}$; $\text{Change} = \frac{10^{-5}}{10^{-4}} = \frac{1}{10}$
    (Decreased by 10 times)

Worksheet – 2

1. **Correct Answer: B**
   (H_2SO_4) is a mineral acid that undergoes complete ionization in aqueous solution, unlike acetic and formic acids (weak organic acids) or HCN (weak mineral acid).
2. **Correct Answer: A**
   $H_2SO_4 \rightarrow 2H^+ + SO_4^{2-}$ (Basicity = 2)
3. **Correct Answer: B**
   CH_3COOH undergoes partial ionization in water.
4. **Correct Answer: C**
   Ammonium hydroxide is a weak base because it only partially dissociates into its constituent ions in an aqueous solution.
5. **Correct Answer: A**
   $NaOH \rightarrow Na^+ + OH^-$ (Acidity = 1)
6. **Correct Answer: B**
   Bases are chemically characterized by a sharp, bitter taste and a slippery/soapy feel, which serves as a sensory cue to their fundamental chemical properties.
7. **Correct Answer: B**
   (CH_3COOH) is classified as a weak acid because it only partially ionizes in aqueous solution, establishing an equilibrium where the majority of molecules remain undissociated.
8. **Correct Answer: C**
   $Al(OH)_3 \rightarrow Al^{3+} + 3OH^-$ Acidity = 3.
9. **Correct Answer: C**
   Sodium hydroxide (NaOH) is a strong Arrhenius base that undergoes complete dissociation in aqueous solution.
10. **Correct Answer: A**
    $Ba(OH)_2 \rightarrow Ba^{2+} + 2OH^-$ (Acidity = 2)

Worksheet – 3

1. **Correct Answer: D**
   HNO_3 is a strong acid that dissociates completely to release H^+, whereas CO_2, SO_3, and $FeCl_3$ act as Lewis acids but lack ionizable hydrogen.
2. **Correct Answer: B**
   $H_2PO_2^-$ (hypophosphite ion) is derived from H_3PO_2, which is a monoprotic acid where the remaining two hydrogens are bonded directly to phosphorus (P-H) and cannot be donated.
3. **Correct Answer: D**
   The reaction between hydronium and hydroxide is a Brønsted-Lowry neutralization because it involves proton transfer (H^+ from H_3O^+ to OH^-) and a Lewis neutralization because the OH^- ion donates an electron pair to the proton.
4. **Correct Answer: D**
   The pair $H_2PO_4^-$ and PO_4^{3-} differs by two protons, making it an invalid pair.
5. **Correct Answer: A**
   $\text{Base} + H^+ \rightarrow \text{Conjugate Acid}$;
   $O^{2-} + H^+ \rightarrow OH^-$
6. **Correct Answer: D**
   HN_3 (Hydrazoic acid) is an acid that dissociates to release H^+ and the azide ion.
7. **Correct Answer: B**
   The ammonium ion (NH_4^+) is a Brønsted-Lowry acid because it is a proton donor, but as a cation, it does not fit the classical Arrhenius definition.
8. **Correct Answer: C**
   Aqueous SO_3 forms sulphuric acid (H_2SO_4), which increases the hydronium ion concentration in water, satisfying the Arrhenius definition of an acid.
9. **Correct Answer: D**
   Acetylene is the weakest acid in the set, its conjugate base (the acetylide ion) is the strongest.
10. **Correct Answer: A**
    HNO_3 is a strong acid that lacks the ability to accept a proton in standard aqueous condition.

Worksheet – 4

1. **Correct Answer: C**
   Na_2KPO_4 is a mixed salt as it contains two different basic radicals: Na^+ and K^+
2. **Correct Answer: B**
   Plaster of Paris undergoes a hydration reaction when mixed with water, re-forming gypsum, which is a hard, crystalline solid mass.
3. **Correct Answer: B**
   Glauber's salt ($Na_2SO_4 \cdot 10H_2O$), spontaneously lose their water of crystallization to the atmosphere when the vapor pressure of the hydrate exceeds the partial pressure of water vapor in the surrounding air.
4. **Correct Answer: D**
   $NaAlO_2$ is a highly soluble inorganic salt.
5. **Correct Answer: D**
   Passing chlorine gas over dry slaked lime yields bleaching powder, which is chemically represented as calcium oxychloride
6. **Correct Answer: C**
   Because both the cation and anion are negligible conjugate partners that do not undergo significant hydrolysis in water.
7. **Correct Answer: B**
   Washing soda is a basic salt because it is derived from a strong base (NaOH) and a weak acid (H_2CO_3), resulting in an aqueous solution with pH > 7 due to the hydrolysis of the carbonate ion.
8. **Correct Answer: D**
   $NaHCO_3 + H^+ \rightarrow Na^+ + H_2O + CO_2(g)\uparrow$

9. **Correct Answer: C**

Plaster of Paris is chemically known as calcium sulphate hemihydrate, meaning it contains half a molecule of water of crystallization for every molecule of calcium sulphate.

10. **Correct Answer: D**

$CaCO_3 \xrightarrow{\Delta} CaO + CO_2 \uparrow$

Competitive Corner

SINGLE CORRECT OPTION TYPE

1. **Correct Answer: B**

Basicity = number of replaceable H^+ ions per molecule.

2. **Correct Answer: C**

Sulphuric acid (H_2SO_4) is known as the "King of Chemicals" due to its indispensable role in various industrial processes and its massive global production volume.

3. **Correct Answer: B**

Methyl orange is a pH indicator that undergoes a structural change (quinonoid transformation) in acidic conditions $pH < 3.1), resulting in a red color.

4. **Correct Answer: A**

$[H^+]_{new} = \frac{[H^+]_{initial}}{1000} \Rightarrow$

$[H^+]_{new} = 10^{-9} M$; $pH_{new} = -\log_{10}(10^{-9})$ = 9

5. **Correct Answer: A**

Neutralization is the exothermic reaction where H^+ ions from an acid and OH^- ions from a base combine to form H_2O and a salt.

6. **Correct Answer: A**

Acetic acid is a differentiating solvent with a lower dielectric constant and lower proton affinity than water, which leads to incomplete ionization of electrolytes that would otherwise be strong in water.

7. **Correct Answer: D**

While water, ammonia, and alcohols are standard amphiprotic solvents (capable of both donating and accepting protons), acetic acid is primarily classified as a protogenic (acidic) solvent in most comparative chemical frameworks.

8. **Correct Answer: C**

BCl_3 is a classic Lewis acid due to its vacant p-orbital, but it is not a Brønsted acid because it lacks a proton (H^+) to donate.

9. **Correct Answer: C**

A Lewis acid-base reaction involves the donation of a lone pair from the base to the acid, forming a coordinate covalent (dative) bond.

10. **Correct Answer: D**

Strength follows pK_a values: HCO_3^- (10.3) < HSO_4^- (1.9) < H_3O^+ (1.7) < HSO_3F (-15)

11. **Correct Answer: C**

$Al(OH)_3 \rightleftharpoons Al^{3+} + 3OH^-$ (Acidity = 3)

12. **Correct Answer: D**

$K_w = [H^+][OH^-]$; since $[H^+] = [OH^-]$, then $[H^+] = \sqrt{K_w}$

13. **Correct Answer: B**

The industrial synthesis of NaOH involves the electrolysis of brine (saturated aqueous NaCl), where chloride ions are oxidized at the anode and water is reduced at the cathode, leaving NaOH in the electrolyte.

14. **Correct Answer: A**

$2NaCl(aq) + 2H_2O(l) \xrightarrow{\text{electrolysis}} 2NaOH(aq) + Cl_2(g) + H_2(g)$

15. **Correct Answer: A**

Chemical Formula of Gypsum: $CaSO_4 \cdot 2H_2O$

16. **Correct Answer: C**

Ammonium hydroxide is unstable and decomposes on warming to give ammonia (NH_3) and water (H_2O).

17. **Correct Answer: B**

When a metal reacts with an acid, the metal displaces hydrogen from the acid, so hydrogen gas (H_2) is evolved.

ONE OR MORE THAN ONE CORRECT OPTION TYPE

18. Correct Answer: A, B, D

While Arrhenius and Brønsted-Lowry theories quantify strength via dissociation constants (K_a/K_b) and proton-transfer equilibrium, Lewis theory lacks a universal scale for strength because adduct stability is highly dependent on the specific reference partner (Hard-Soft Acid-Base interactions). Arrhenius theory explains acids and bases only in aqueous solutions by defining acids as substances that increase H^+ concentration and bases as substances that increase OH^- concentration.

19. Correct Answer: A, C, D

$BeCl_2$ is predominantly covalent (Fajans' Rules) and acts as a Lewis acid due to the vacant 2p orbitals in the valence shell of Be, not the second shell of an "atom" in general or due to ionic character. $BeCl_2$ does not donate a proton (H^+), so it is not a Brønsted acid.

20. Correct Answer: A and D

The autoprotolysis of water is an endothermic process, therefore, increasing temperature shifts the equilibrium to the right, decreasing pH.

$$-\log[H^+] = -\log(K_w^{1/2}) \Rightarrow pH = \frac{1}{2}pK_w$$

21. Correct Answer: A, B and C

The pH (concentration of H^+) is an intensive property dependent on the equilibrium constant (affected by temperature), the chemical identity of the species (Nature), and the extent of ionization (degree of dissociation), but it is independent of the total Volume of the solution.

22. Correct Answer: A, B and D

$$CO_2 + H_2O \rightarrow H_2CO_3,$$

$$SO_3 + H_2O \rightarrow H_2SO_4,$$

$$N_2O_5 + H_2O \rightarrow 2HNO_3$$

23. Correct Answer: B, C and D

Al^{3+} undergoes hydrolysis producing H^+. Salt of strong acid (HCl) and strong base (NaOH). Salt of weak base (NH_3) and strong acid (HCl).

ASSERTION AND REASON TYPE

24. Correct Answer: A

The increase in the number of electronegative oxygen atoms and the higher oxidation state of Sulfur in SO_3 (+6 vs +4 in SO_2) significantly enhances the electrophilicity of the sulfur center, facilitating the acceptance of an electron pair

25. Correct Answer: A

$$HCl(aq) \rightarrow H^+(aq) + Cl^-(aq);$$

$$\text{Acid} \rightarrow \text{Proton} + \text{Conjugate Base}$$

26. Correct Answer: C

Adding water *will* restore the blue color of the complex.

27. Correct Answer: A

The high enthalpy of hydration for concentrated acids makes the dilution process intensely exothermic.

28. Correct Answer: A

Ethanoic acid is a weak acid with a low degree of dissociation, leading to a lower H^+ concentration and a significantly slower rate of hydrogen evolution when reacting with magnesium.

MATRIX MATCHING TYPE

29. Correct Answer:

a-S, b-R, c-Q, d-P , Washing soda: Na_2CO_3. $10H_2O$, Blue vitriol: $CuSO_4.5H_2O$,

Baking Soda: $NaHCO_3$, Gypsum: $CaSO_4.2H_2O$

30. Correct Answer:

a-S, b-P, c-Q, d-R, Sodium Family: Na_2CO_3, Magnesium Family: $Mg(OH)_2$

Chloride Family: KCl, Sulphate Family: K_2SO_4

COMPREHENSION TYPE

Passage – I

31. Correct Answer: C

Citric acid acts as a sequestrant and acidulant in carbonated beverages, providing a sharp "tangy" flavor profile.

32. Correct Answer: C

Mineral (inorganic) acids are typically derived from geological mineral sources and are characterized by the absence of carbon-hydrogen bonds, unlike organic acids which are metabolic products of biological organisms.

33. Correct Answer: C

Tartaric acid (2,3-dihydroxybutanedioic acid) is a dicarboxylic acid found naturally in tamarind.

Passage – II

34. Correct Answer: A.

$$pK_b = -\log_{10}(K_b)$$

$$\text{Higher } pK_b \Rightarrow \text{Lower } K_b \Rightarrow \text{Weaker Base}$$

35. Correct Answer: D

For a weak acid, the equilibrium of the protolysis reaction lies heavily to the left, resulting in a low acid dissociation constant (K_a) because the concentration of the undissociated species [HA] remains high relative to the concentration of the conjugate base [A^-] and hydronium ions [H_3O^+].

36. Correct Answer: C

Strong acids are characterized by virtually complete ionization in aqueous solution ($\alpha \approx 1$), which mathematically results in an exceptionally large K_a value and a correspondingly low (often negative) pK_a value, representing a high thermodynamic drive for proton donation.

INTEGER TYPE

37. Correct Answer: 6

The chemical formula of Epsom salt is $MgSO_4.7H_2O$

38. Correct Answer:

Bleaching powder is written as $CaOCl_2$ (or Ca(OCl)Cl), which contains two chlorine atoms.

39. Correct Answer: 0.3

$$N_{mix} = \frac{N_1V_1 + N_2V_2}{V_1 + V_2}$$

$$pH = -\log[H^+] = -\log(0.5) \approx 0.301$$

Previous Year Questions

1. Correct Answer: B

Hydroxide ion concentration is logarithmic to pOH, which is inversely related to pH (pOH = 14 - pH); thus, the solution with the highest pH (P) has the highest OH^-.

2. Correct Answer: C

The dehydration of Gypsum (Y) to Plaster of Paris (X) occurs at a critical temperature of 373 K (100°C) to yield the hemihydrate; exceeding this temperature leads to "dead burnt plaster" ($CaSO_4$).

3. Correct Answer: A

Only Experiments I (Slaked lime, a base) and III (Vinegar, an acid) show correct color transitions for all four indicators; Experiment II is incorrect (Methyl Orange turns red in acid), and Experiment IV is incorrect (Phenolphthalein turns pink in base).

4. Correct Answer: B

The bulb illuminates only when solution X is an electrolyte (acid or base) that provides mobile charge carriers; glucose is a non-electrolyte, and solid NaCl lacks ion mobility.

5. Correct Answer: C

In the Chlor-alkali process, chlorine evolves at the anode (X) and hydrogen at the cathode (Y); NaOH is formed near the cathode, not the anode, and the process produces caustic soda, not baking soda.

6. Correct Answer: B

Thermolysis of Sodium hydrogen carbonate ($NaHCO_3$) produces Sodium carbonate (a washing powder constituent) and CO_2 gas (which precipitates $CaCO_3$ in lime water).

7. Correct Answer: C

These pH values match the characteristic ranges of the listed biological and household fluids, where lemon juice is strongly acidic and milk of magnesia is moderately basic.

8. Correct Answer: C

The reaction describes the protonation of the bicarbonate anion, acting as a Brønsted-Lowry base, to form carbonic acid which subsequently undergoes rapid dehydration to evolve CO_2 gas (Y); this endothermic decomposition and gas evolution mechanism is the primary basis for its application in soda-acid fire extinguishers and systemic antacids.

9. Correct Answer: D

This flowchart delineates the Solvay process where the carbonation of ammoniacal brine (i) yields sparingly soluble sodium bicarbonate (ii) due to the common-ion effect; subsequent thermal calcination produces sodium carbonate (iii) and regenerates CO_2 (iv), which is analytically confirmed via the formation of a $CaCO_3$ precipitate (V) in a saturated $Ca(OH)_2$ solution.

10. Correct Answer: C

Potassium acetate (CH_3COOK) is the salt of a weak acid (CH_3COOH) and a strong base (KOH), leading to significant anionic hydrolysis where the acetate ion acts as a conjugate base to generate OH^- ions, thereby driving the equilibrium pOH lower and the pH above 7.

5. Chemical bonding

Solutions

Worksheet – 1

1. **Correct Answer: A**
   Explanation: electrons present in the outermost shell of an atom and are responsible for chemical bonding and reactivity
2. **Correct Answer: D**
   Explanation:
   Formal charge
   = (Valence electrons in free atom)
   (Non-bonding electrons + ½ Bonding electrons)
3. **Correct Answer: C**
   Explanation: Formal charge = (Valence electrons) - (Non-bonding electrons + ½ Bonding electrons)
   = 7 - (0 + ½ × 8); = 7 – 4; = +3
4. **Correct Answer: B**
   Explanation: C (Group 14): 4 valence electrons, S (Group 16): 6 valence electrons
   F (Group 17): 7 valence electrons
5. **Correct Answer: D**
   Explanation : CH_4 (Methane): Carbon has 8 electrons in valence shell → octet satisfied.
   CO_2: Carbon and Oxygen both satisfy octet rule.
6. **Correct Answer: A**
   Explanation: Combining capacity = number of bonds an atom can form = Valency.
7. **Correct Answer: B**
   Explanation: In NH_3, nitrogen forms 3 bonds with hydrogen.
   Hence, valency of N = 3.
8. **Correct Answer: C**
   Explanation: CO, NH_3, H_2O, CH_4 all follow octet rule.
9. **Correct Answer: D**
   Explanation: CH_4 (Carbon has exactly 8 electrons) → not super octet.
10. **Correct Answer: C**
    Explanation: In CO_2, carbon forms 2 double bonds → total 4 bonds.
    Valency = number of bonds

Worksheet-2

1. **Correct Answer: A**
   Explanation: A) Kessel and Lewis
2. **Correct Answer: C**
   Explanation:
   - Chlorine typically gains 1 electron to complete its octet → forming Cl^- ion with 8 electrons in its outer shell.
   - Neutral Cl atom has 7 valence electrons → not fully stable.
   - Cl^+ and Cl^{2+} ions are uncommon and less stable.
3. **Correct Answer: C**
   Explanation:
   - Oxygen (Group 16) has **6** valence electrons.
   - In Lewis structure, these 6 electrons are represented as dots around the oxygen symbol.
4. **Correct Answer: C**
   Explanation:
   - Lewis symbols show only valence electrons, which participate in bonding.
5. **Correct Answer: C**
   Explanation:
   - Fluorine atoms share 1 pair of electrons → single bond.
   - Fluorine is a halogen with 7 valence electrons, forming one single bond to complete octet.

6. **Correct Answer: B**
   Explanation:
   - Nitrogen in NH_3 has one lone pair of electrons.
   - Hydrogen atoms do not have lone pairs.
7. **Correct Answer: C**
   Explanation:
   - H_2 and Cl_2 have single bonds.
   - NH_3 has single bonds between N and H.
   - O_2 molecule has a double bond between the two oxygen atoms.
8. **Correct Answer: C**
   Explanation:
   - Oxygen has 6 valence electrons.
   - In water, oxygen forms 2 single bonds with 2 hydrogen atoms (2 bonds × 2 electrons = 4 electrons).
   - Remaining 4 valence electrons (2 pairs) stay as two lone pairs on oxygen.
   - So, oxygen has two lone pairs in H_2O.
9. **Correct Answer: C**
   Explanation:
   - Carbon has 4 valence electrons.
   - In methane, carbon shares one electron with each of 4 hydrogen atoms forming four single covalent bonds.
   - Carbon completes its octet with these 4 bonds.
10. **Correct Answer: C**

    Explanation:
    - Lewis dots represent the number of valence electrons:
    - Be (Group 2): 2 valence electrons → 2 dots
    - B (Group 13): 3 valence electrons → 3 dots
    - Li (Group 1): 1 valence electron → 1 dot
    - N (Group 15): 5 valence electrons → 5 dots

Worksheet-3

1. **Correct Answer: A**

   Explanation**:** When atoms form a chemical bond, they achieve a more stable arrangement by sharing or transferring electrons.
   - Stability in chemistry is often associated with lower potential energy.
   - Atoms tend to combine in ways that minimize the total potential energy of the system.
   - Hence, forming a chemical bond releases energy, which means the potential energy of the system decreases.
2. **Correct Answer: B**

   Explanation: Ionic compounds are formed by attraction between positive and negative ions.

   This attraction is an electrostatic force (Coulombic attraction between oppositely charged ions).
3. **Correct Answer: B**

   Explanation**:** In KCl, potassium (K) loses one electron to form K^+ ion → electrovalency is +1.
4. **Correct Answer: B**

   Explanation**:** Ionic compounds form between metal and non-metal.
5. **Correct Answer: A**

   Explanation: Ionic bond strength increases with high charge on ions (greater attraction).

   Smaller cations and larger anions favor strong ionic bonding because smaller cation can approach closely and large anion reduces repulsion.
6. **Correct Answer**: **C**

   Explanation**:** NaCl, KCl, $MgCl_2$ are ionic compounds.

   H_2O is a covalent compound (sharing electrons).

7. **Correct Answer: D**
   Explanation : Covalency refers to the number of electrons an atom **shares** in covalent bonds.
8. **Correct Answer: C**
   Explanation**:** In CO_2, carbon forms two double bonds with oxygen atoms → 4 bonds in total.
   Covalency = number of shared electrons = 4.
9. **Correct Answer: D**
   Explanation : NH_3 has a lone pair on nitrogen; BF_3 is electron deficient and can accept lone pairs.
   The bond formed when lone pair is donated to an empty orbital → coordinate covalent bond (also called dative bond).
10. **Correct Answer: B**
    Explanation: A coordinate covalent bond (also called a dative bond) is formed when both electrons in the shared pair come from the same atom.
    Unlike a regular covalent bond where each atom contributes one electron, here one atom donates both electrons, and the other atom accepts them.
    - Once formed, the coordinate bond behaves like a normal covalent bond (electrons are shared), but the origin of the electrons is different.
    - Hydrogen bond is a different type of intermolecular force, so option C is unrelated.
    - Electrons are shared, so D is incorrect.

Worksheet-4

1. **Correct Answer: A) P and P orbitals**

   Explanation: p-p > s-p > s-s

   Directional Nature**:** p-orbitals are directional (dumbbell-shaped), whereas s-orbitals are non-directional (spherical). Because p-orbitals are concentrated along a specific axis, they can overlap much more effectively than s-orbitals.

2. **Correct Answer: B) Orbitals overlap will not occur**
   Explanation:
   This is governed by the Pauli Exclusion Principle and the requirements for covalent bonding.
   - Pauli Exclusion Principle**:** This principle states that no two electrons in an atom (or a shared molecular orbital) can have the same set of four quantum numbers. Essentially, if two electrons occupy the same space, they must have opposite spins (↑↓)
   - Repulsion: If two atoms approach each other with electrons of the same spin (↑↑), they will experience strong inter-electronic repulsion. Instead of a stable bond forming, the potential energy of the system increases, and the atoms repel each other.
3. **Correct Answer: C**

   Explanation

   p-p orbitals overlapping "end-to-end" (axially) perfectly describes the formation of a σ - bond.

4. **Correct Answer: B) sp-sp**

   Explanation: The efficiency of overlapping depends on the s-character of the hybrid orbital. An sp orbital has 50% s-character, sp^2 has 33.3%, and sp^3 has 25%. Because s-orbitals are closer to the nucleus and more compact, a higher s-character leads to a shorter, more directional, and stronger bond.
   Therefore, sp-sp overlap is the most efficient.

5. **Correct Answer: B) BF_3**

   Explanation: Boron in BF_3 has 3 valence electrons and forms 3 sigma bonds with Fluorine atoms with no lone pairs. This corresponds to a steric number of 3, which results in trigonal planar geometry and sp^2 hybridization.

6. **Correct Answer: B) One sigma and two pi bonds**
   Explanation: Calcium carbide is an ionic compound consisting of Ca^{2+} and the carbide ion $[C \equiv C]^{2-}$. The two carbon atoms are held together by a triple bond, which consists of one σ (sigma) bond and two π (pi) bonds.
7. **Correct Answer: B) sp^2 hybridisation**
   Explanation**:** In ethylene (CH_2=CH_2), each carbon atom is bonded to two hydrogens (2 sigma bonds) and one carbon (1 sigma bond and 1 pi bond). Since each carbon has 3 sigma bonds and no lone pairs, it adopts sp^2 hybridization to form a trigonal planar structure around each carbon.
8. **Correct Answer: B) $sp^3 > sp^2 > sp$**
   Explanation: The size of a hybrid orbital is inversely proportional to its s-character.
   - sp^3: 75% p-character (Largest/most elongated)
   - sp^2: 66.7% p-character
   - sp: 50% p-character (Smallest/most compact)

   As s-character increases, the electron density is pulled closer to the nucleus, making the orbital smaller.
9. **Correct Answer: D) Increases progressively**
   Explanation: The bond angles for these hybridization states are:
   sp^3: 109.5° (Tetrahedral);sp^2: 120° (Trigonal Planar);sp: 180° (Linear)
   As the state changes from sp^3 to sp, the angle increases.
10. **Correct Answer: A) sp^3d**

    Explanation: Phosphorus has 5 valence electrons. In PCl_5, it forms 5 sigma bonds with 5 Chlorine atoms and has zero lone pairs.

    Steric Number = 5 (sigma bonds) + 0 (lone pairs) = 5

    A steric number of 5 corresponds to sp^3d hybridization, resulting in a trigonal bipyramidal geometry.

Worksheet-5

1. **Correct Answer: B) Trigonal planar, sp^2**
   - Explanation: Boron (B) has 3 valence electrons and forms 3 sigma bonds with three Fluorine (F) atoms. Since there are no lone pairs on the central atom, the steric number is 3 (3 bond pairs + 0 lone pairs).
   - A steric number of 3 corresponds to sp^2 hybridisation, which results in a trigonal planar geometry with bond angles of 120°
2. **Correct Answer: B) BF_3, AlF_3**
   - Explanation: Both Boron (B) and Aluminum (Al) belong to Group 13 and have 3 valence electrons.
   - In both BF_3 and AlF_3, the central atom forms 3 sigma bonds with no lone pairs, leading to sp^2 hybridisation and trigonal planar geometry for both.
3. **Correct Answer: B) Octahedral**
   - Explanation**:** sp^3d^2 hybridisation involves the mixing of one s, three p, and two d orbitals, totaling 6 hybrid orbitals.
   - To minimize repulsion, these six orbitals point toward the corners of a regular octahedron, resulting in octahedral geometry with 90° bond angles.
4. **Correct Answer: D) Octahedral**
   - Explanation: Sulfur (S) has 6 valence electrons and forms 6 sigma bonds with six Fluorine atoms in SF_6.
   - With 6 bond pairs and 0 lone pairs, the steric number is 6, requiring sp^3d^2 hybridisation. This results in an octahedral shape.

5. **Correct Answer: D) Pentagonal bipyramidal**
   - Explanation**:** Iodine (I) has 7 valence electrons and forms 7 sigma bonds with seven Fluorine atoms.
   - With a steric number of 7 (7 bond pairs + 0 lone pairs), the central atom undergoes sp^3d^3 hybridisation.
6. **Correct Answer: B) B (Boron)**
   Explanation: A 120° bond angle indicates trigonal planar geometry and sp^2 hybridisation. This requires the central atom to have 3 valence electrons to form 3 bonds, which is characteristic of Boron.
7. **Correct Answer: D)** $CO < CO_2 < CO_3^{2-}$
   Explanation**:** Bond length is inversely proportional to bond order:
   - CO: Triple bond (Bond order = 3) ⟶ Shortest.
   - CO_2 : Double bond (Bond order = 2) ⟶ Intermediate.
   - CO_3^{2-} : Partial single/double bond due to resonance (Bond order ≈ 1.33) ⟶ Longest.
8. **Correct Answer: C) O-O**
   Explanation**:** Single bonds between small, highly electronegative atoms like N, O, and F are weakened by strong lone pair-lone pair repulsions. Among these, the O-O single bond has the lowest bond energy (approx.142 kJ/mol) compared to C-C, N-N, or F-F.
9. **Correct Answer: A) XeF_2,** IF_2^-
   Explanation: Both XeF_2 and IF_2^- have 2 bond pairs and 3 lone pairs on the central atom. According to VSEPR theory, these electron pairs arrange in a trigonal bipyramidal orientation, but the molecular shape (positions of the atoms) is linear for both.
10. **Correct Answer: A) Three**
    Explanation: In a regular octahedron, there are three pairs of opposite vertices. Each pair forms one straight X-M-X line, resulting in exactly three 180° bond angles.

Competitive Corner

SINGLE CORRECT OPTION TYPE

11. **Correct Answer: A) Chemical bond**
    Explanation**:** A chemical bond is the physical process responsible for the attractive interactions between atoms and molecules, which confers stability to diatomic and polyatomic chemical compounds.
12. **Correct Answer: B) Exothermic**
    Explanation**:** Atoms bond together to reach a lower energy state and achieve stability. Because the bonded state is more stable (lower energy) than the individual atoms, energy is released when a bond forms.
13. **Correct Answer: B) Inertness of noble gases**
    Explanation: Kossel and Lewis noticed that noble gases (like Neon and Argon) are very stable and rarely react. They concluded that atoms of other elements enter into chemical bonds to achieve a stable octet (8 electrons in the valence shell), similar to noble gases.
14. **Correct Answer: B) Core electrons**
    Explanation: Valence electrons are in the outermost shell and participate in bonding.
    Core electrons are located in the inner shells and do not typically participate in chemical reactions.
15. **Correct Answer: A) BCl_3**
    Explanation: A "contracted octet" occurs when the central atom has fewer than 8 electrons in its valence shell after bonding.
    - In BCl_3, Boron (B) has 3 valence electrons. It forms 3 single bonds with Chlorine atoms.
    - Total electrons around Boron = 3 + 3 = 6 electrons.
    - Since 6 is less than 8, it is a contracted octet. (Note: SF_6 and PCl_5 are examples of expanded octets).

16. **Correct Answer: A) 0**
    Explanation: The formula for formal charge is:
    FC = [Total valence e^-] - [Non-bonding e^-] - $\frac{1}{2}$ [Bonding e^-]
    - In O=C=O, Carbon has 4 valence electrons.
    - It has 0 non-bonding electrons (lone pairs).
    - It has 8 bonding electrons (4 bonds \times 2).
    - FC = 4 - 0 - $\frac{1}{2}$ (8) = 0.
17. **Correct Answer: B) Loss of 2 electrons**
    Explanation: Magnesium (Mg) is in Group 2 and has an atomic number of 12 (electronic configuration: 2, 8, 2). To achieve a stable octet (like Neon: 2, 8), it loses its 2 outermost electrons to form the Mg^{2+} ion.
18. **Correct Answer: D) Ionic bond**
    Explanation**:** Potassium (K) is a metal that loses one electron, and Chlorine (Cl) is a non-metal that gains one electron. The resulting electrostatic attraction between the K^+ cation and Cl^- anion forms an ionic bond.
19. **Correct Answer: A) Good conductors**
    Explanation**:** In the solid state, ions are locked in a rigid lattice and cannot move. However, when melted (infused) or dissolved in water (aqueous), the lattice breaks down, allowing the ions to move freely and conduct electricity.
20. **Correct Answer: D) RbCl**
    Explanation: Ionic bonds (like the one in Rubidium Chloride) are non-directional because the electrostatic attraction between oppositely charged ions acts in all directions around the ion. Covalent bonds (like in NCl_3, $BeCl_2$, and BCl_3) are directional because they involve specific orbital overlaps at fixed angles.
21. **Correct Answer: D) Al_2O_3 > CaO > $MgBr_2$ > NaCl**
    Explanation**:** Lattice energy depends on two main factors:
    1. Ionic Charge**:** Higher charges lead to higher lattice energy. Al2O3 has the highest charges ($Al^{3+,}$ O^{2-}), followed by CaO (Ca^{2+} ,O^{2-}).
    2. Ionic Size: Smaller ions have higher lattice energy. Comparing CaO and $MgBr_2$, O^{2-} is much smaller than Br^-, giving CaO a higher lattice energy despite Mg^{2+} being small. NaCl has the lowest charges (+1/-1) and thus the lowest lattice energy.
22. **Correct Answer: B) Covalent bond**
    Explanation: A covalent bond is defined by the sharing of electron pairs between atoms to achieve stability, typically allowing each atom to attain a stable outer shell.
23. **Correct Answer: D) F — F**
    Explanation**:** A bond is non-polar when electrons are shared equally between two atoms of the same electronegativity. In F_2, both atoms are identical, so there is no difference in electronegativity, resulting in a zero dipole moment.
24. **Correct Answer: D) Large cation and small anion**
    Explanation: Fajan's rules describe what increases covalent character (small cation, large anion). Therefore, the **ionic** character is favored by the opposite conditions: a large cation (which has low polarizing power) and a small anion (which is hard to polarize).
25. **Correct Answer: D) BF_3**
    Explanation: In Boron Trifluoride (BF_3**)**, the central Boron atom has 3 valence electrons and forms 3 single bonds with no lone pairs. According to VSEPR theory, these 3 bond pairs push apart to a 120° angle, creating a trigonal planar shape.
    - NH_3 and PCl_3 are trigonal pyramidal due to a lone pair.
    - IF_3 is T-shaped.

26. Correct Answer: A) 25% s-character

- Explanation**:** An sp^3 hybrid orbital is formed by mixing one 's' orbital and three 'p' orbitals (total 4 orbitals).
- Percentage of s-character = $\frac{1}{4}$ X 100 = 25%
- Percentage of p-character = $\frac{3}{4}$ X 100 = 75%

27. Correct Answer: D) 4, 3

Explanation: * For NH_4^+ : Nitrogen forms 4 sigma bonds with 0 lone pairs (sp^3 hybridization). Number of hybrid orbitals = 4.

- For NO_3^- : Nitrogen forms 3 sigma bonds (one is a double bond, but we only count the sigma component) with 0 lone pairs (sp^2 hybridization). Number of hybrid orbitals = 3.

28. Correct Answer: C) 1:3

Explanation: A bond angle of $109^\circ 28'$ is characteristic of sp^3 hybridization (tetrahedral). sp^3 has 25% s-character and 75% p-character. The ratio 25:75 simplifies to 1:3.

29. Correct Answer: B) Pyramidal

Explanation: In the Chlorate ion (ClO_3^-), Chlorine is the central atom.

- Valence electrons of Cl (7) + 1 (from negative charge) = 8 electrons.
- It forms 3 bonds with Oxygen and retains one lone pair.
- The electron geometry is tetrahedral, but with one lone pair, the molecular shape becomes pyramidal (similar to NH_3).

30. Correct Answer: D) 4, 0

Explanation: In the Lewis structure of the Nitrate ion (NO_3^-), Nitrogen is bonded to three Oxygen atoms. To satisfy the octet rule, it forms one double bond and two single bonds (one of which is a coordinate bond).

- Total shared pairs (bond pairs) = 4 (1 double + 2 single).
- Nitrogen uses all its valence electrons for these bonds, leaving **0** lone pairs.

ONE OR MORE THAN ONE CORRECT OPTION TYPE

31. Correct Answer: D) Electron pair repulsion determines shape

Explanation**:** The core premise of VSEPR theory is that valence electron pairs surrounding an atom repel each other, and the molecule adopts a geometry that minimizes this repulsion.

32. Correct Answers: A) CH_4, B) NH_3, C) H_2O

Explanation: All three molecules have 4 electron domains (bond pairs + lone pairs) around the central atom, requiring sp^3 hybridization. BF_3 is sp^2 hybridized.

33. Correct Answers: A) Number of valence electrons, B) Ability to lose, gain, or share electrons, D) Electronic configuration of the valence shell

Explanation: Valency is the combining capacity of an atom, determined primarily by the number of electrons in its outermost (valence) shell and how many it must lose, gain, or share to reach a stable configuration.

34. Correct Answers: A) NaCl, B) MgO, D) CaF_2

Explanation: These are formed by metals reacting with non-metals through electron transfer. HCl is primarily considered a polar covalent compound in the gaseous state.

35. **Correct Answers: A) Small electronegativity difference, B) High ionization energy, C) Sharing tendency of atoms**
Explanation: Covalent bonds are favored when atoms have similar electronegativities (small difference), preventing complete electron transfer. High ionization energy makes it difficult to remove electrons to form ions, leading atoms to share instead

ASSERTION AND REASON TYPE

36. **Correct Answer: a) Both Assertion (A) and Reason (R) are true and R is the correct explanation of A**
Explanation:
The reason explains the assertion because the presence of that lone pair is exactly what allows the specific mechanism of "single-atom contribution" to occur.
37. **Correct Answer: a) Both Assertion (A) and Reason (R) are true and R is the correct explanation of A**
Explanation:
The reason correctly explains the assertion: the increased attraction (R) is the physical cause that results in the shorter distance (A). For example, a C≡C triple bond is much shorter than a C-C single bond.
38. **Correct Answer: a) Both Assertion (A) and Reason (R) are true and R is the correct explanation of A**
Explanation: Multiple bonds (double and triple) share more electron pairs than single bonds, which creates a stronger electrostatic "glue" pulling the atoms together. This increased attraction makes the bond harder to break, resulting in higher bond energy.
39. **Correct Answer: a) Both Assertion (A) and Reason (R) are true and R is the correct explanation of A**
Explanation: Electronegativity measures how strongly an atom attracts shared electrons. A larger difference between two atoms means the more electronegative atom pulls the electrons closer to itself, leading to an unequal distribution of charge (polarity).
40. **Correct Answer: d) Assertion (A) is false but Reason (R) is true**
Explanation:
    - Assertion (A) is False: Lone pairs do affect hybridization because they occupy hybrid orbitals just like bonding pairs do. For example, Nitrogen in NH_3 is sp_3 hybridized specifically because it has 3 bonding pairs and 1 lone pair (total 4 domains).
    - Reason (R) is True: Hybridization is determined by the "steric number," which is the total count of electron domains (lone pairs + bonding groups) around the central atom.

MATRIX MATCHING TYPE

41. **(a)sp Hybridization (Linear)**
    - How it forms**:** One s orbital mixes with one p orbital.
    - Shape: The two hybrid orbitals sit opposite each other to minimize repulsion.
    - Example**:** $BeCl_2$ or C_2H_2 (Ethyne).
    - Angle: 180°.

    (b). sp^2 Hybridization (Trigonal Planar)
    - How it forms: One s orbital mixes with two p orbitals.
    - Shape: The three orbitals point toward the corners of an equilateral triangle.
    - Example: BF_3 or C_2H_4 (Ethene).
    - Angle**:** 120° .

    (c). sp^3 Hybridization (Tetrahedral)
    - How it forms: One s orbital mixes with all three p orbitals.
    - Shape: A 3D shape where four orbitals point to the corners of a tetrahedron.
    - Example: CH_4 (Methane).
    - Angle: 109.5°

    (d). sp^3d Hybridization (Trigonal Bipyramidal)

- How it forms: One s, three p, and one d orbital mix.
- Shape: It has two types of positions: three equatorial (forming a triangle) and two axial (pointing up and down).
- Example: PCl_5.
- Angles: 120° (between equatorial) and 90° (between axial and equatorial).

(II)

(a) – Q

Bond length is the distance between nuclei; it gets shorter for double/triple bonds.

(b) – R

Bond angle is the spatial angle; it increases as s-character increases

(sp = 180° vs sp^3 = 109.5°).

(c) – S

Bond enthalpy is the "breaking energy"; higher energy means a much stronger, more stable bond.

(d) – P

Bond order is the number of bonds; as it goes up (1 \to 2 \to 3), bond length goes down.

COMPREHENSION TYPE

Passage – I

42. Correct Answer: B

Explanation:

To find the total number of valence electrons, we add the valence electrons of each atom in the molecule:

- Nitrogen (N): Belongs to Group 15, so it has 5 valence electrons.
- Hydrogen (H): Belongs to Group 1, so each has 1 valence electron. Since there are 3 Hydrogen atoms: 3 x 1 = 3 valence electrons.
- Total: 5 + 3 = 8 valence electrons.

43. Correct Answer: B) One lone pair and three bond pairs

Explanation:

When we draw the Lewis structure for NH_3

1. Nitrogen is the central atom.
2. It forms three single covalent bonds with the three Hydrogen atoms (using 6 electrons total).
3. The remaining 2 electrons (from the total of 8) stay on the Nitrogen atom as a non-bonding pair, also known as a lone pair.
4. Therefore, Nitrogen has 3 bond pairs and 1 lone pair.

44. Correct Answer: B) All atoms in NH_3 achieve a stable configuration

Explanation:

Nitrogen achieves an octet (8 electrons), and each Hydrogen achieves a duplet (2 electrons), which is the stable configuration for the first shell.

Passage – II

45. Correct Answer: B) Tetrahedral

Explanation**:** Electron geometry considers all regions of electron density (both bonding pairs and lone pairs) around the central atom.

46. Correct Answer: C) Trigonal pyramidal

Explanation**:** Molecular shape describes the arrangement of the atoms only, even though the lone pair still influences the position of those atoms.

47. Correct Answer: B) Lone pair–bond pair repulsion is greater than bond pair–bond pair repulsion

Explanation:

In a perfect tetrahedral molecule (like Methane, CH_4), the bond angle is 109.5°.

INTEGER TYPE

48. Correct Answer: The formal charge on nitrogen is 0.

Explanation: FC = 5 - 2 - $\frac{5}{2}$;FC = 5 - 2 - 3 = 0

49. Correct Answer: 1

Explanation: Valency is the combining capacity of an atom. Hydrogen has 1 electron in its outer shell and needs 1 more to achieve a stable "duplet" configuration (like Helium). Therefore, it typically forms one bond.

50. Correct Answer: sp^2

Explanation: Central Atom**:** Nitrogen (N)
Bonded Atoms: 3 Oxygen atoms.
Lone Pairs: In NO_3^- nitrogen has no lone pairs (it uses all its valence electrons for bonding to satisfy the formal charges of the oxygens).
Steric Number**:** 3 (atoms)} + 0 (lone pairs)} = 3.

51. Correct Answer: 109.5°

Explanation: Methane consists of a central Carbon atom bonded to four Hydrogen atoms. According to VSEPR theory, these four electron pairs repel each other equally to be as far apart as possible in 3D space.

- Geometry: Tetrahedral
- Bond Angle: Because all four bonds are identical and there are no lone pairs to cause distortion, the bond angle is perfectly symmetrical.

Previous year question paper

1. Correct Answer: A

CH_3-$\underline{CH_2}$-$CH=CH_2$: The underlined carbon is bonded to two Carbons and two Hydrogens via single bonds. This is sp^3.

B) CH_3-$\underline{CH_2}$-NH_2: Similar to above, this carbon has 4 sigma bonds. This is also sp^3.

D) CH_3-$\underline{CH_2}$-CN: The underlined carbon has 4 sigma bonds. This is sp^3.

2. Correct Answer: C

O_2^- (1.5) + N_2 (3) + O_2^+ (2.5) = 7

3. Correct Answer: A

NH_3 to NH_4^+: NH_3 is sp^3 (3 bond pairs + 1 lone pair). NH_4^+ is sp^3 (4 bond pairs). No change.

4. Correct Answer: Total: 5 molecules.
Correct Option: C

- ClF_3: 10 electrons around Cl. (Yes)
- SF_6: 12 electrons around S. (Yes)
- CH_4: 8 electrons around C. (No)
- NH_3: 8 electrons around N. (No)
- SF_4: 10 electrons around S. (Yes)
- XeF_4: 12 electrons around Xe. (Yes)
- PCl_5: 10 electrons around P. (Yes)

5. Correct Answer: D (0 and -1)

The structure of Ozone is a resonance hybrid. In a single Lewis structure:

- The central oxygen has a formal charge of +1.
- One terminal oxygen (double bonded) has a charge of 0.
- The other terminal oxygen (single bonded) has a charge of -1.

6. Correct Answer: A

i. SF_6 is sp^3d hybridised: Incorrect**.**
SF6 is sp^3d^2 (octahedral).

ii. $S_2O_3^{2-}$ is a linear ion: Incorrect**.**
Thiosulfate is tetrahedral around the central sulfur.

iii. There is no pi-bonding in SO_4^{2-} ion: Incorrect. SO_4^{2-} has $d\pi - p\pi$ bonding.

7. Correct Answer: B

sing Molecular Orbital Theory (Total electrons):

1. NO^+ (14 electrons): Same as N_2, Bond Order = 3.0
2. NO (15 electrons**):** Bond Order = 2.5
3. NO^- (16 electrons): Bond Order = 2.0

8. Correct Answer: B (sp^2, sp)

Based on the structure provided in the prompt (H_3C-$CH=C=CH_2$):

- C_2**:** Involved in one double bond (CH=). Hybridization = sp^2
- C_3**:** Central carbon of the allene system bonded to two double bonds (=C=). Hybridization = sp

9. Correct Answer: B) H_2O, $BeCl_2$, BF_3:

- H_2O: Oxygen has **2 lone pairs** that repel the bond pairs strongly, pushing them closer together.(104.5°)
- BF_3 : Boron has **0 lone pairs** and 3 bonding pairs that spread out equally in a flat plane.(120°)
- $BeCl_2$: Beryllium has **0 lone pairs** and only 2 bonding pairs, which stay as far apart as possible (180°)
- *Order:* $104.5° < 120^\circ ° < 180^\circ$ (This is increasing order).

10. Correct Answer: C

According to Valence Bond Theory (VBT) for coordination compounds:

1. sp^3: Involves one 4s and three 4p orbitals (e.g., in tetrahedral complexes like $[NiCl_4]^{2-}$.
2. dsp^2: Involves one inner 3d, one 4s, and two 4p orbitals (e.g., in square planar complexes like $[Ni(CN)_4]^{2-}$). d^2sp^3: Involves two inner 3d, one 4s, and three 4p orbitals (e.g., in inner-orbital octahedral complexes like $[NH_3)_6]^{3+}$ While sp^3d^2 (outer-orbital octahedral) is also possible using 4d orbitals, the options provided specifically highlight the use of the 3d (inner) and 4s (outer) orbitals.

6. General Organic Chemistry

Solutions

Worksheet – 1

1. **Correct Answer: B**
   Urea exhibits the covalent catenation and functional group chemistry characteristic of organic systems.
2. **Correct Answer: D**
   Friedrich Wöhler's 1828 synthesis is the pivotal experiment that disproved Berzelius's "Vital Force Theory" by producing organic urea from purely inorganic ammonium cyanate via thermal rearrangement.
3. **Correct Answer: A**
   Alicyclic (Aliphatic Cyclic) compounds are ring systems that lack aromatic character
4. **Correct Answer: A**
   C_2: Double-bonded carbon → sp^2, C_4: Saturated methylene ($-CH_2-$) → sp^3, C_5: Triple-bonded carbon → sp
5. **Correct Answer: D**
   The common name is derived from the Latin word for ant, *formica*, as the acid was historically isolated from the distillation of red ants
6. **Correct Answer: D**
   Carbon-14 (^{14}C) is a cosmogenic radionuclide formed from ^{14}N that decays via beta-emissio ($^{14}C \rightarrow {}^{14}N$) after an organism dies
7. **Correct Answer: A**
   Proteins (polyamides of amino acids) are classic organic biopolymers.
8. **Correct Answer: A**
   Diamond and graphite are allotropes of carbon because they are different structural forms of the same element in the same physical state.
9. **Correct Answer: B**
   Non-benzenoid aromatics typically refers to aromatic carbocycles that do not contain a benzene ring.
10. **Correct Answer: C**
    C_2: double bonded carbon → sp^2,
    C_4: single bonded carbon → sp^3

Worksheet – 2

1. **Correct Answer: B**
   Saponification is the alkaline hydrolysis of triglycerides (triesters). The byproducts are soap and glycerol.
2. **Correct Answer: D**
   Cleaning soaps are defined as the water-soluble sodium (hard soap) or potassium (soft soap) salts of long-chain fatty acids.
3. **Correct Answer: D**
   The π-electron cloud in ethene acts as a nucleophile, allowing electrophiles to add across the double bond, converting it to a saturated system.
4. **Correct Answer: B**
   The reaction of carboxylic acid with bicarbonate ion produces water and carbon dioxide gas.
5. **Correct Answer: C**
   Unlike soaps (carboxylates), synthetic detergents are typically anionic surfactants based on long-chain alkylbenzene sulphonates. General Structure: $R\text{-}SO_3^-Na^+$
6. **Correct Answer: D**
   Soap. Saponification results in the formation of soap and glycerol.
7. **Correct Answer: A**
   Soft soaps (liquid soaps/shaving creams) are potassium salts of fatty acids formed using Caustic Potash (KOH).

8. **Correct Answer: D**
Addition reactions are characteristic of unsaturated systems possessing accessible π-electron clouds.
9. **Correct Answer: A**
Sodium acetate is formed on reaction of acetic acid with sodium carbonate.
10. **Correct Answer: D**
Soap molecules are amphiphiles. They consist of a long, non-polar hydrocarbon chain (tail) that interacts with oils via London dispersion forces (Hydrophobic), and a polar ionic carboxylate group ($-COO^{-}Na^{+}$) that is known as hydrophilic head.

Worksheet – 3

1. **Correct Answer: B**
According to IUPAC nomenclature, C_5 → pent
2. **Correct Answer: D**
The principal chain contains 6 carbons (hexane), ruling out pentane derivatives. The structure lacks double lines, indicating an alkane (suffix "-ane"), eliminating options B and C ("-ene"). Numbering from left-to-right yields the locant set {2, 3, 4}, whereas right-to-left yields {3, 4, 5}. The rule of lowest locants dictates the former is correct.
3. **Correct Answer: A**
The name implies a 5-carbon parent chain with two methyl groups at C_2 (quaternary carbon) and one methyl group at C_4 (tertiary carbon).
4. **Correct Answer: D**
The longest continuous carbon chain is identified as containing 8 carbons (octane). Numbering from left-to-right yields locants 3 and 4 for the substituents. Numbering from right-to-left would yield 5 and 6. The lowest locant set rule prioritizes the 3,4 combination. While the methyl group is at position 3 and ethyl at 4, IUPAC rules dictate that substituents are listed alphabetically in the final name (Ethyl before Methyl).
5. **Correct Answer: D**
In an internal alkyne like 2-hexyne, the geometry around the triple bond forces the two sp-carbons and the two alpha-carbons directly attached to them to lie on the same axis, creating a linear sequence of four carbon atoms.
6. **Correct Answer: C**
The double bond gets priority for the lowest number, so numbering starts from the right. The longest chain containing the alkene has 4 carbons (Butene). Two methyl groups are located at C_3.
7. **Correct Answer: A**
The chain is numbered to give the lowest possible set of locants to the multiple bonds, regardless of type. The suffix "ene" always precedes "yne" alphabetically. The terminal "e" of "ene" is dropped if followed by a vowel or "y"
8. **Correct Answer: B**
Esters are named as alkyl alkanoates.
9. **Correct Answer: C**
The longest continuous carbon chain consists of 4 carbons (Butane). Numbering from either side yields methyl substituents at positions 2 and 3. Two methyl groups will be named as dimethyl.
10. **Correct Answer: B**
The longest chain is propane parent backbone. The central carbon acts as a quaternary center bonded to two additional methyl groups.

Worksheet – 4

1. **Correct Answer: C**

The carbonyl group is divalent, allowing for different alkyl group combinations (e.g., Methyl/Propyl vs. Ethyl/Ethyl).

2. **Correct Answer: D**

Iodine is a monovalent halogen. It attaches to a single carbon and cannot serve as a bridge between alkyl groups; thus, it can only form position isomers.

3. **Correct Answer: D**
Allyl alcohol is an acyclic unsaturated alcohol. Cyclopropanol is a cyclic saturated alcohol.
4. **Correct Answer: A**
Because they possess fundamentally different functional groups, they are classified as functional isomers.
5. **Correct Answer: D**
Both possess the same molecular formula but they differ in connectivity.
6. **Correct Answer: C**
A homologue series for alkynes requires the presence of at least one carbon-carbon triple bond.
7. **Correct Answer: D**
By definition, a homologous series is a family of compounds with the same general formula and functional group, where each successive member differs by one methylene unit (-CH_2-) in the carbon chain.
8. **Correct Answer: D**
Metamers in esters arise from the different distribution of alkyl groups around the central ester linkage.
9. **Correct Answer: B**
Acetone and propanal share the same molecular formula but contain different functional groups.
10. **Correct Answer: D**
Optical isomerism falls under stereoisomerism.

Competitive Corner

SINGLE CORRECT OPTION TYPE

1. **Correct Answer: D**

Friedrich Wöhler synthesized Urea from inorganic Ammonium Cyanate in 1828.

2. **Correct Answer: B**

The number of sp^3 hybrid orbits of carbon are 12 and the number of s-orbitals of hydrogen are 10.

3. **Correct Answer: C**

The total sigma bond count is the sum of C-C and C-H bonds.

4. **Correct Answer: D**

"Hydrocarbon" is the fundamental parent class for compounds composed exclusively of Carbon and Hydrogen.

5. **Correct Answer: B**

Formic acid consists of a carbonyl group bonded to a hydrogen atom and a hydroxyl group.

6. **Correct Answer: D**

The compound contains a Carbon-Carbon triple bond, the suffix for that is “-yne”

7. **Correct Answer: C**

The principal functional group is the Alcohol. The longest chain is 5 carbons long containing the functional group. This leaves the ethyl and methyl group as substituents.

8. **Correct Answer: A**

An alkyl group is formed by the removal of one hydrogen atom from its corresponding alkane.

9. **Correct Answer: B**

Chain isomers have the same molecular formula but different carbon skeletons. S-butyl (4-carbon chain) and t-butyl (3-carbon chain)

10. **Correct Answer: A**

The -OH group is on a primary carbon in isobutyl alcohol and The -OH group is on the tertiary central carbon in the t-butyl alcohol.

11. **Correct Answer: C**

n-hexane, 2-methylpentane, 3-methylpentane, 2,2-dimethylbutane, and 2,3-dimethylbutane are the constitutional isomers of the hexane.

12. **Correct Answer: A**

In a tie for lowest locants, Alkene > Alkyne

13. **Correct Answer: A**

because the ketone functional group shifts its position relative to the chain ends.

14. **Correct Answer: C**

Carbon 3 is bonded to C-2, C-4 and a methyl group. Carbon 5 is bonded to C-4, C-6 and a methyl group.

15. **Correct Answer: B**

The hydrophilic ionic head group of the soap molecule is stabilized in the polar solvent (water) through electrostatic attraction between its ionic charge and the permanent dipoles of the water molecules.

16. **Correct Answer: B**

At 443 K concentrated sulfuric acid acts as a powerful protonating and dehydrating agent, facilitating the β-elimination of a water molecule from ethanol to form ethene.

17. **Correct Answer: C**

A blue (non-sooty) flame is formed during complete combustion when sufficient oxygen is available. This leads to efficient burning, no soot formation, and a higher temperature compared to a sooty flame, which results from incomplete combustion.

18. **Correct Answer: B**

On heating ethanol at 443 K with concentrated H_2SO_4, ethanol undergoes dehydration, eliminating a molecule of water to form ethene (an unsaturated hydrocarbon).

19. **Correct Answer: B**

Alkaline $KMnO_4$ or acidified $K_2Cr_2O_7$ act as oxidising agents. They add oxygen to ethanol during heating, converting it into ethanoic acid.

20. **Correct Answer: B**

In a substitution reaction, one atom or group of atoms replaces another. Here, chlorine replaces a hydrogen atom in methane in the presence of sunlight, forming chloromethane (CH_3Cl) and HCl.

21. **Correct Answer: B**

Temporary hardness is due to calcium and magnesium bicarbonates $[Ca(HCO_3)_2, Mg(HCO_3)_2]$. On boiling, these decompose to form insoluble carbonates, removing the hardness.

22. **Correct Answer: C**

Higher bond multiplicity means more shared electrons, leading to stronger attraction between nuclei and electrons, hence a shorter bond length.

ONE OR MORE THAN ONE CORRECT OPTION TYPE

23. **Correct Answer: A, B, D**

(C_1-C_2) is a terminal alkyne triple bond, which consists of 1 σ and 2 π bonds. C_3-C_2 is the single bond connecting the alkyne and alkene groups. A single bond consists of 1 σ and 0 π bonds. C_3-C_4 is an alkene double bond which consists of 1 σ and 1 π bond.

24. **Correct Answer: A, C**

Alicyclic heterocyclic compounds contain a ring with at least one non-carbon atom (heteroatom) but do not possess aromatic character (they behave like aliphatic compounds).

25. **Correct Answer: A and B**

The parent chain is wrongly numbered. Because the parent chain was numbered incorrectly, the positions (locants) assigned to the Bromine and Chlorine atoms are also incorrect.

26. **Correct Answer: A, B, C**

The molecule exhibits isomerism. This molecule can be carboxylic acid, ester or hydroxy alkene

27. **Correct Answer: B and C**

In option B, the position of the triple bond is different in each compound. In the option C, each compound has different functional group.

28. **Correct Answer: A, B, C, D**
Tautomerism involves the dynamic equilibrium between two structural isomers (usually via proton shift).

29. **Correct Answer: B, D**
Tautomerism requires the presence of at least one acidic α-hydrogen on an sp^3 carbon adjacent to the carbonyl group.

30. **Correct Answer: B**
Option B consists of ester and ketone + ether. Functional isomers share the same molecular formula but possess different functional groups.

ASSERTION AND REASON TYPE

31. **Correct Answer: A**
The Reason gives the correct general formula. So both the Assertion and Reason are true, and the Reason correctly explains the Assertion.

32. **Correct Answer: D**
n-Propyl alcohol (propan-1-ol) and isopropyl alcohol (propan-2-ol) have the same functional group (–OH) and differ only in its position, so they are position isomers, not functional isomers.

33. **Correct Answer: A**
The Reason correctly defines unsaturated hydrocarbons as those having at least one C=C or C≡C bond. Since ethane has none, it is saturated.

34. **Correct Answer: A**
The presence of α-hydrogens is essential for keto–enol tautomerism, so 2-pentanone can exist in equilibrium with its enol form. Hence, both the Assertion and the Reason are true, and the Reason correctly explains the Assertion.

35. **Correct Answer: A**
In 2-butyne ($CH_3–C≡C–CH_3$), the two terminal methyl ($–CH_3$) carbons each form four σ bonds (three C–H and one C–C), so they are sp^3 hybridized. The two carbons involved in the triple bond are sp hybridized. Hence, exactly two sp^3 carbons are present.

MATRIX MATCHING TYPE

36. **Correct Answer:**
a–P, b–R, c–Q, d–S
   a) All C–C and C–H bonds are single; both carbons are sp^3.
   b) Contains a C≡C triple bond; both carbons are sp.
   c) As an alkene (e.g., propene), it contains a C=C bond → sp^2 carbons.
   d) Toluene-type structure: aromatic ring carbons are sp^2, methyl carbon is sp^3.

37. **Correct Answer:**
a–Q, b–P, c–R, d–S
   a) Presence of a carbon–carbon triple bond.
   b) Contains the –OH functional group.
   c) Contains the >C=O group within the carbon chain.
   d) Contains the –CHO functional group.

COMPREHENSION TYPE

38. **Correct Answer: A**
The carbon of the –COOH (carboxyl) group is attached to only one other carbon, so it is a primary (1°) carbon.

39. **Correct Answer: C**
Carbon 1 ($CH_3–$) is attached to only one carbon → 1°. Carbon 2 is attached to four carbons → 4°

40. **Correct Answer: C**
C_3 is bonded to three carbon atoms, so it is tertiary (3°)

41. **Correct Answer: C**
Both have the same carbon skeleton and functional group (C=C), but the position of the double bond differs (1-ene vs 2-ene).

42. Correct Answer: A

Alcohols (–OH) and ethers (–O–) can have the same molecular formula but different functional groups, so they are functional isomers.

43. Correct Answer: C

Primary, secondary, and tertiary amines with the same molecular formula are classified as functional isomers because the functional group environment differs.

INTEGER TYPE

44. Correct Answer: 4

Each carbon atom in diamond is tetrahedrally bonded to four other carbon atoms through strong covalent bonds.

45. Correct Answer: 4

In sp^3 hybridisation, four hybrid orbitals are formed, and each makes one sigma bond. Hence, the carbon forms four σ bonds.

46. Correct Answer: 20

Buckminsterfullerene (C_{60}) contains carbon atoms arranged in 20 hexagons and 12 pentagons, giving it a football-like structure.

Previous Year Questions

1. Correct Answer: D

Methanol and ethanol react with sodium metal to evolve hydrogen gas (H_2), not CO_2. Both methanol and ethanol contain the same functional group (–OH), so they do have the same functional group.

2. Correct Answer: B, C and D

–COOH group contains one C=O and one C–O bond (plus C–O to OH), CH_3COOH has three C–H bonds in the CH_3 group and There is one C–C bond (CH_3–CO) and two C–O single bonds.

3. Correct Answer: B

Reacts with ethanol to give a sweet-smelling compound.

4. Correct Answer: B

Ethene burns to give CO_2 and H_2O.

5. Correct Answer: D

Y is ethanoic acid and Ethanoic acid is weaker than sulphuric acid.

6. Correct Answer: B

Chlorofluorocarbons (CFCs) are the main chemicals responsible for ozone layer depletion because they release chlorine radicals in the stratosphere. Chlorine radicals and nitrous oxide (N_2O) also participate in reactions that destroy ozone.

7. Correct Answer: C

(vi) Propanone and (vii) Butanone represent the same homologous series. (ii) Ethanoic acid reacts with ethanol in the presence of an acid catalyst to produce an ester.

8. Correct Answer: D

When ethanol is heated with hot concentrated sulphuric acid (≈443 K), it undergoes dehydration to form ethene (C_2H_4).